Airframe Technology Master

Design

Lecture 28

Flight Controls Systems II

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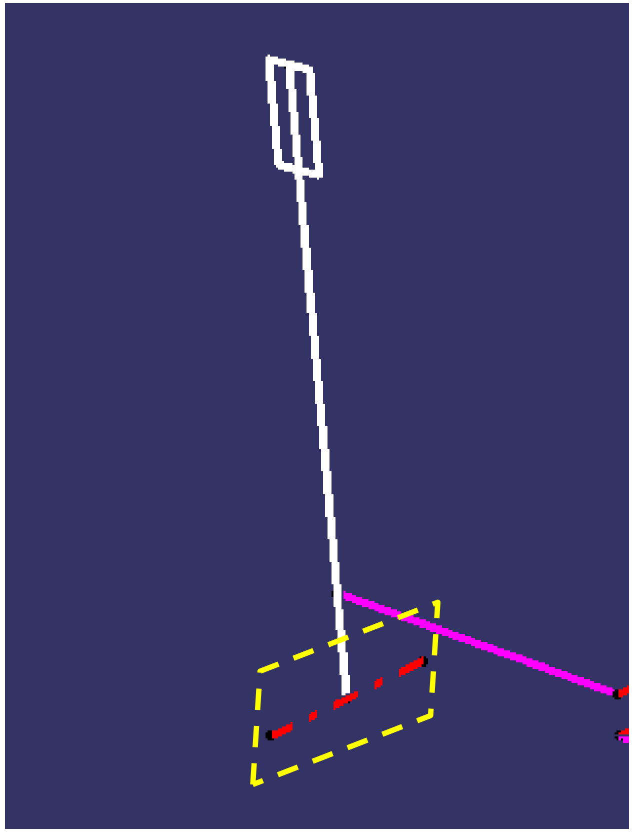
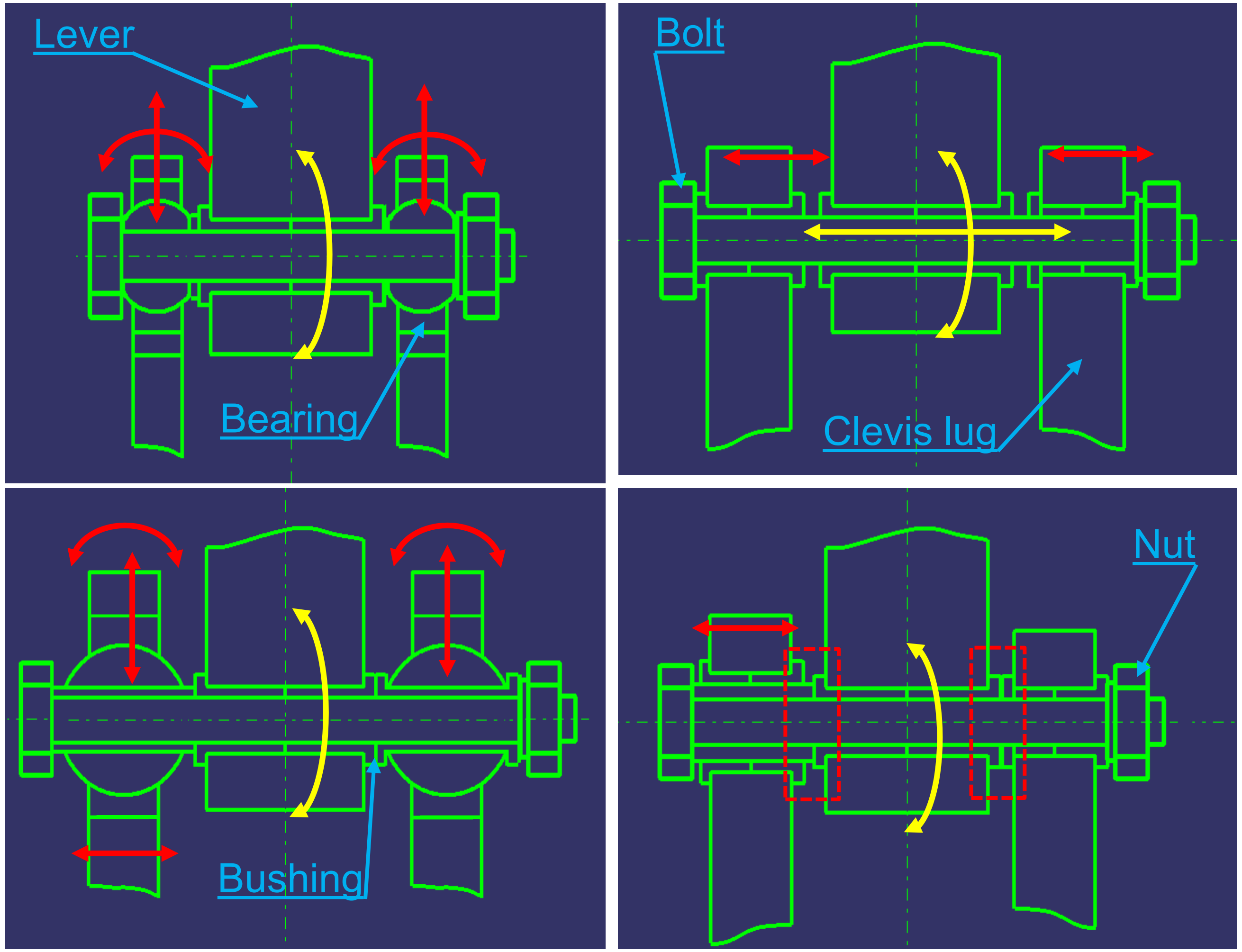


Joint evaluation

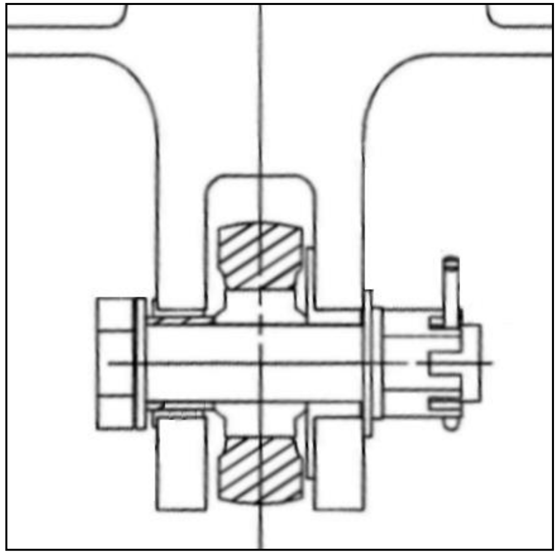
Operational movements

Misalignments

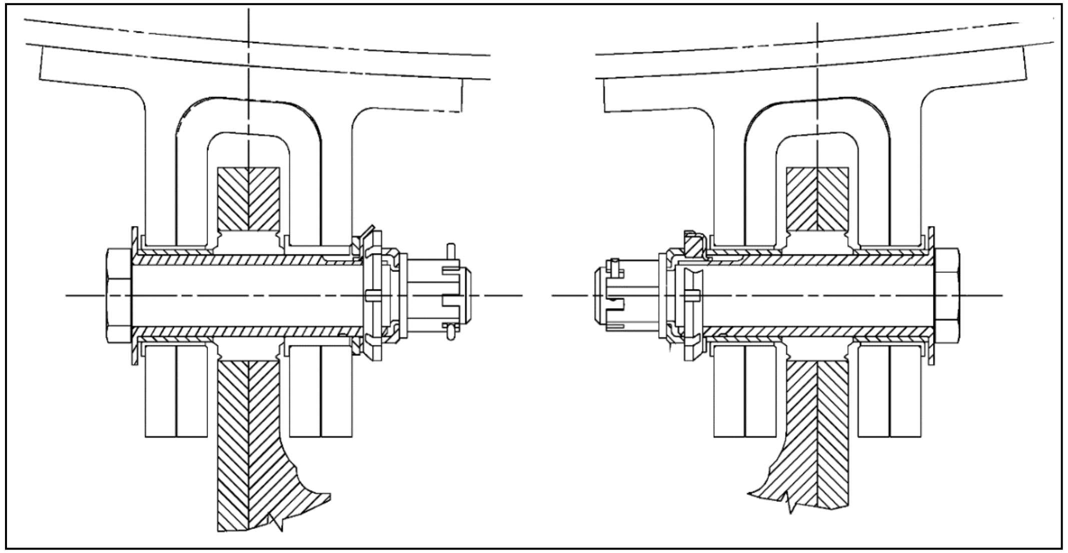


Joint evaluation

Load paths



Single load path

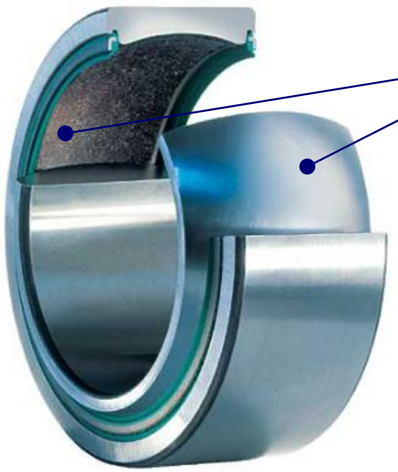


Fail safe

Bearings – Slip paths

Slip paths within rotational joints are the result of relative movement between components within the joint.

- Primary slip paths: within rotational joints are slip paths which are active during normal operation. Primary slip paths should generally be within the bearing.
- Secondary slip paths: within rotational joints are slip paths which are passive during normal operation but may become active when the primary slip path fails (seizes)

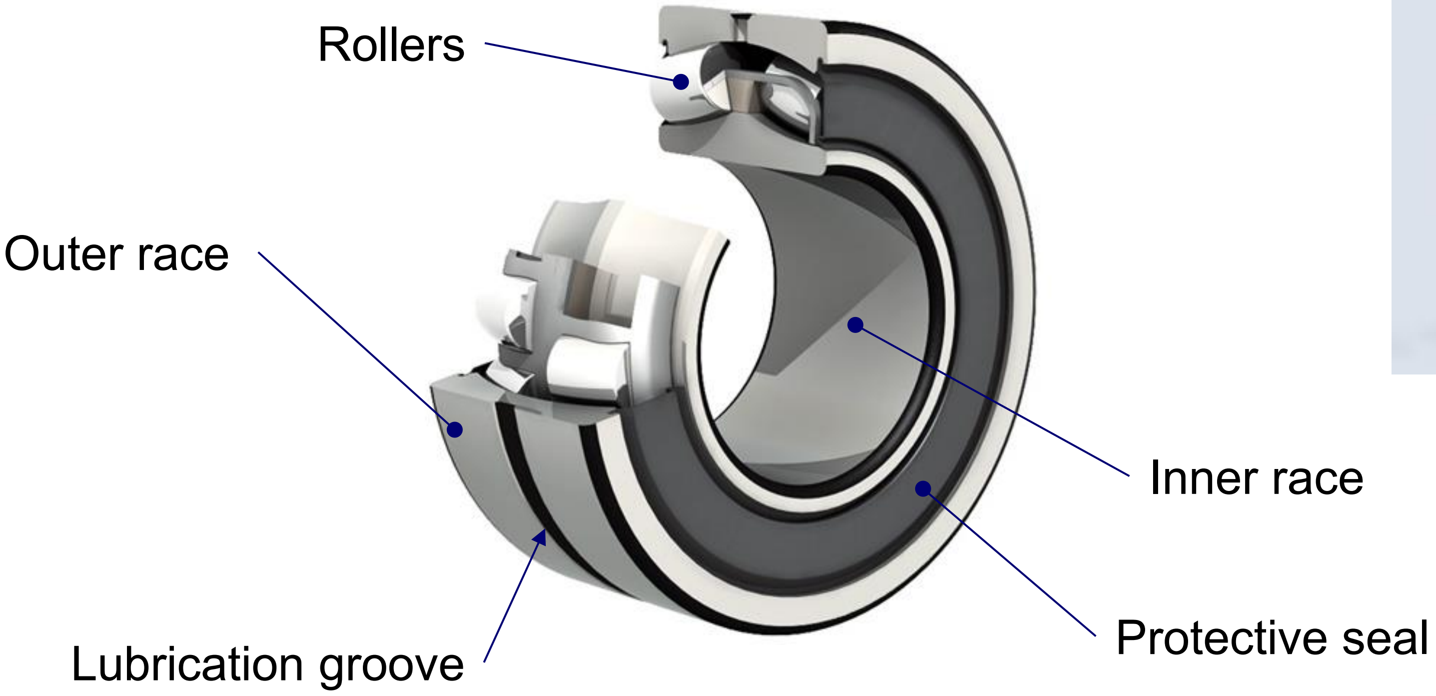


Slip path inside of a plain bearing

Bearings

Rolling bearings – self aligning

- Rolling element bearings comprise an inner and outer race separated by a single or double row of ball or roller bearings. Self-aligning bearings feature a spherical surface on the rolling surface of inner and/or outer races to accommodate misalignment between the bearing housing and the shaft (pin).



Bearings

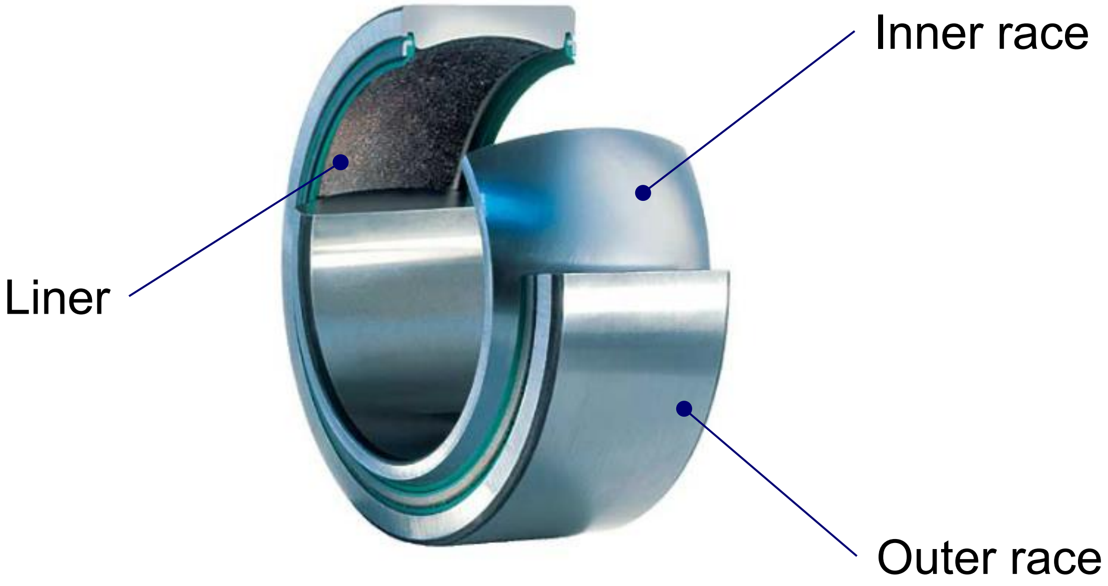
Rolling bearings – self aligning

- Key Features:
  - Very low friction torque.
    - → High load capacity.
    - → Large space envelope and high weight for given pin diameter and duty cycle
    - → Can be protected against contamination using shields or seals.
    - → Low wear rate if correctly lubricated
    - → Small rotational angles (i.e. less than 25°) may cause lubrication problems.
    - → Secondary slip path not generally required, as total seizure is highly unlikely
- Key Considerations:
  - Space envelope.
  - Rotational cycle.
  - Loads (axial, radial, etc)
  - Lubrication.
- Typical Applications:
  - Joints with a large rotational movement.
  - Highly loaded joints.

Bearings

Spherical bearings – self lubricated

- Spherical self-lubricated bearings comprise an inner and outer race each with a corresponding spherical surface. The spherical surface of the outer race is fitted with a low friction liner (typically PTFE). Self-lubricated bearings are initially manufactured with zero clearance between the inner race (ball) and the liner.



Bearings

Spherical bearings – self lubricated

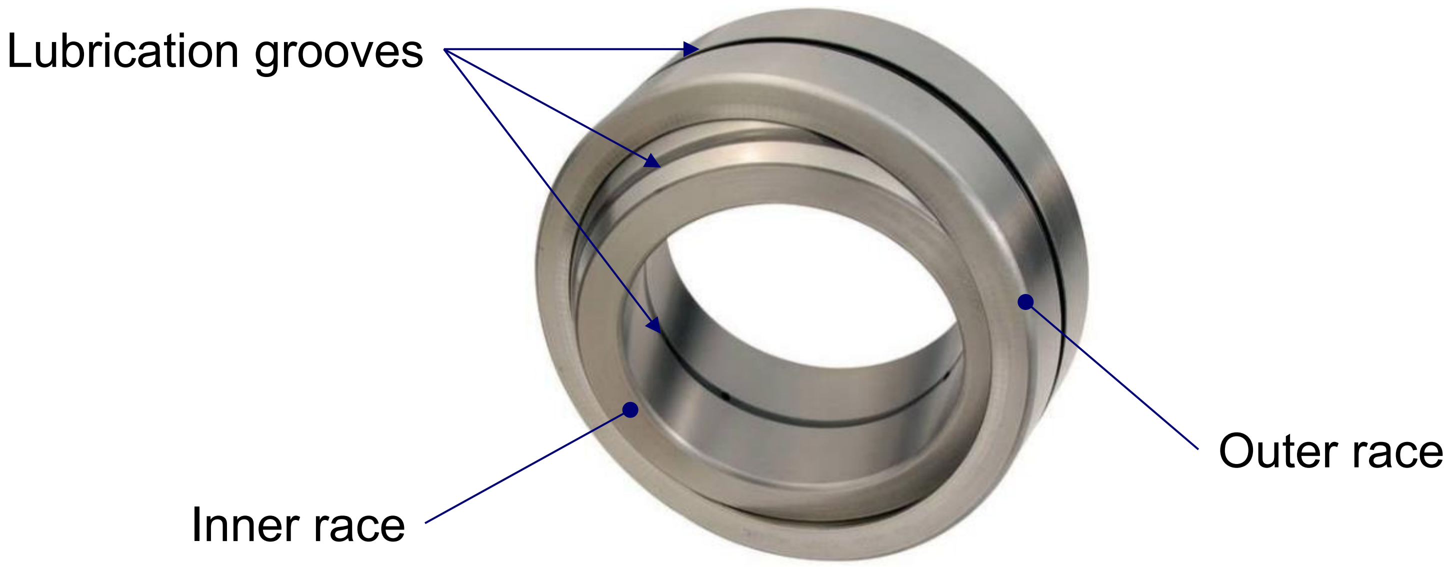
- Key Features:
  - Relatively high friction torque.
  - Moderate load capacity.
  - Small space envelope and low weight for given pin diameter and duty cycle.
  - Moderate wear rate sensitive to environmental conditions and rotational cycle.
  - Does not require lubrication.
  - Secondary slip path may be required in event of seizure.
- Key Considerations
  - Duty cycle.
  - Environment.
  - Friction torque.
  - Reliability of qualification testing due to sensitivity to installation and environmental conditions:
- Typical Applications:
  - Joints with a large rotational movement.
  - Low to moderately loaded joints.

Component installation requirements

Bearings

Spherical bearings – Metal to Metal bearings

- Spherical metal-to-metal bearings comprise an inner and outer race each with a corresponding spherical surface
- The spherical surfaces of the inner and outer races shall be lubricated with grease to prevent wear.
- Spherical metal-to-metal bearings feature a small clearance between the inner race (ball) and the outer race.



Component installation requirements

Bearings

Spherical bearings – Metal to Metal bearings

- Key Features:
  - Moderate friction torque.
  - High load capacity.
  - Small space envelope and low weight for given pin diameter and duty cycle.
  - Low wear rate if correctly lubricated. Requires frequent re-lubrication.
  - Grease transfer holes and grooves introduce stress concentrations.
  - Secondary slip path may be required in event of seizure.
- Key Considerations:
  - Duty cycle.
  - Materials and protective treatments.
  - Lubrication.
  - Backlash.
- Typical Applications:
  - Static joints or joints with a small rotational movement.
  - Highly loaded joints.

Bearings - Installation

General requirements

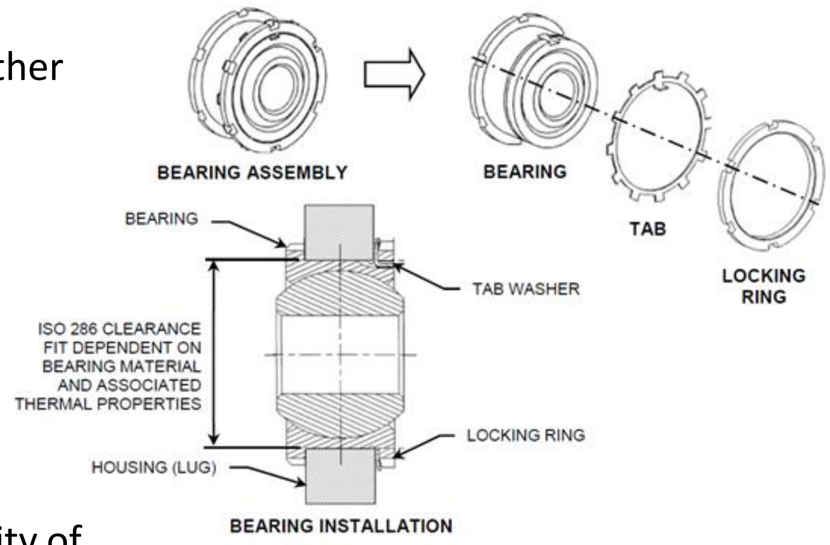
- Spherical bearings or ball bearings shall be easily replaceable without damaging their supports or disassembling any permanently fixed structural attachment
- Non cadmium-plated bearings shall be installed in light alloy supports with interposition sealant and an external sealant. Care must be taken where a grease transfer groove is machined in the outside diameter of the bearing outer race in order to prevent contamination and blockage with sealant.
- Staking on the bearing support shall not be used (spherical marks, notches)

Ball stakes

Bearings - Installation

Retention methods - Mechanical retention (preferred)

- Key Features:.
  - Relatively large and heavy compared with other methods.
  - High axial load capability.
  - Bearing replacement is easier.
- Key Considerations:
  - Bearing friction torque is not affected.
  - Space envelope.
  - Locking of retaining nut.
- Typical Applications:
  - Hinges where axial loads exceed the capability of other retention methods.
  - Joints where axial movement of the bearing in either direction could lead to joint disconnection.
  - Applications where easy bearing replacement is required.
  - Suitable for all bearing types.



Bearings - Installation

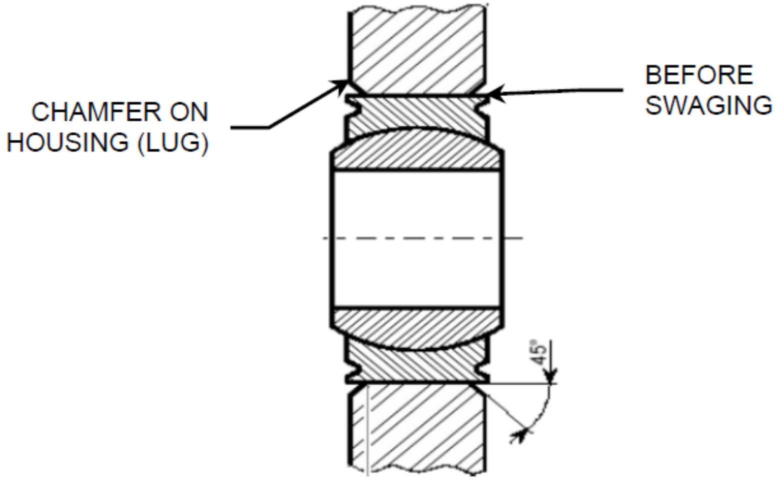
Retention methods - Interference Fit

- Key Features:
  - Lightweight compared with mechanically retained and swaged bush installations.
  - Lower axial load capability (dependent on interference fit).
- Key Considerations:
  - Not suitable for small section rolling element bearings.
  - Housing should be easily removable from the aircraft for bearing replacement.
  - May be used in conjunction with mechanical retention.
  - Installation drawing should specify axial proof load which should be applied in both directions.
- Typical Applications:
  - Medium and heavy duty rolling element bearings.
  - Shouldered and plain fixed bushes

Bearings - Installation

Retention methods - Swaged Bearing

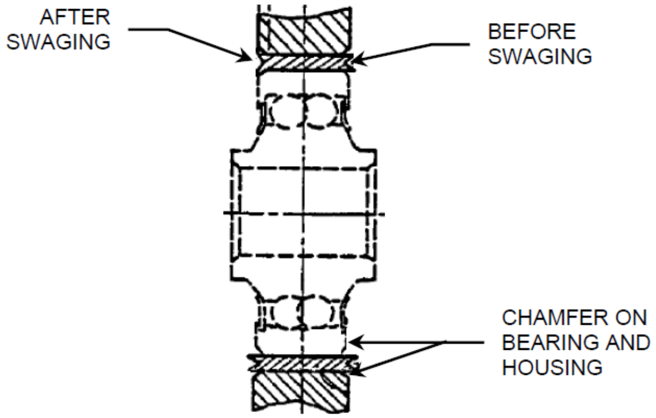
- Key Features:
  - V-groove in the side faces of outer race provides a lip which is swaged onto a chamfer on the housing.
  - Relatively light compared with clamped and swaged bush installations.
- Key Considerations:
  - Bearing friction torque may increase after swaging and requires checking.
  - Housing should be easily removable from the aircraft for bearing replacement.
  - Access for swaging.
  - Installation drawing should specify axial proof load which should be applied in both directions.
- Typical Applications:
  - Spherical bearings.



Bearings - Installation

Retention methods - Swaged intermediate bushing

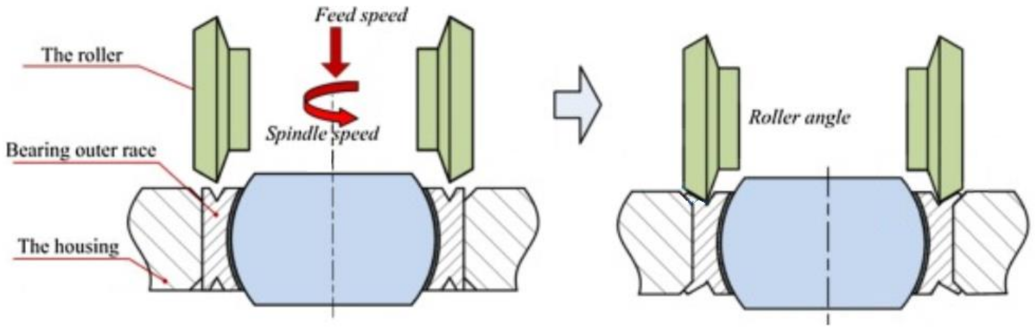
- Key Features:
  - Intermediate V-grooved bush installed between bearing outer race and housing.
  - Bush is an interference fit in the housing and reamed after installation.
  - Swaging operation has no impact on bearing friction torque.
  - Increased housing outside diameter and weight.
- Key Considerations:
  - Space envelope.
  - Access for swaging.
  - Installation drawing should specify axial proof load which should be applied in both directions.
- Typical Applications:
  - All bearing types.

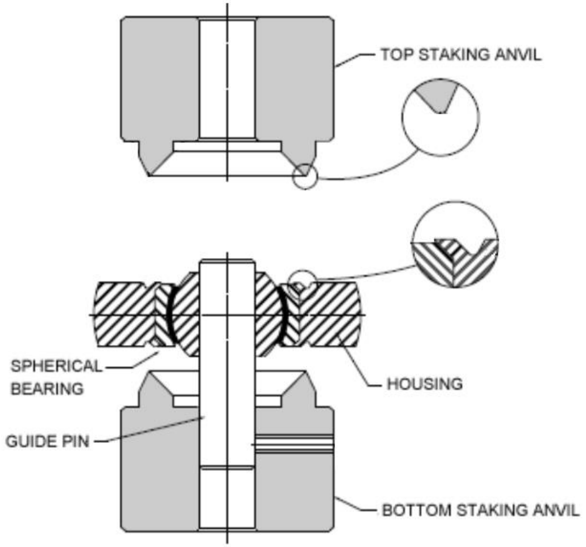


Bearings - Installation

Retention methods – Swaged - Installation tooling

Enough space must be provided for installation



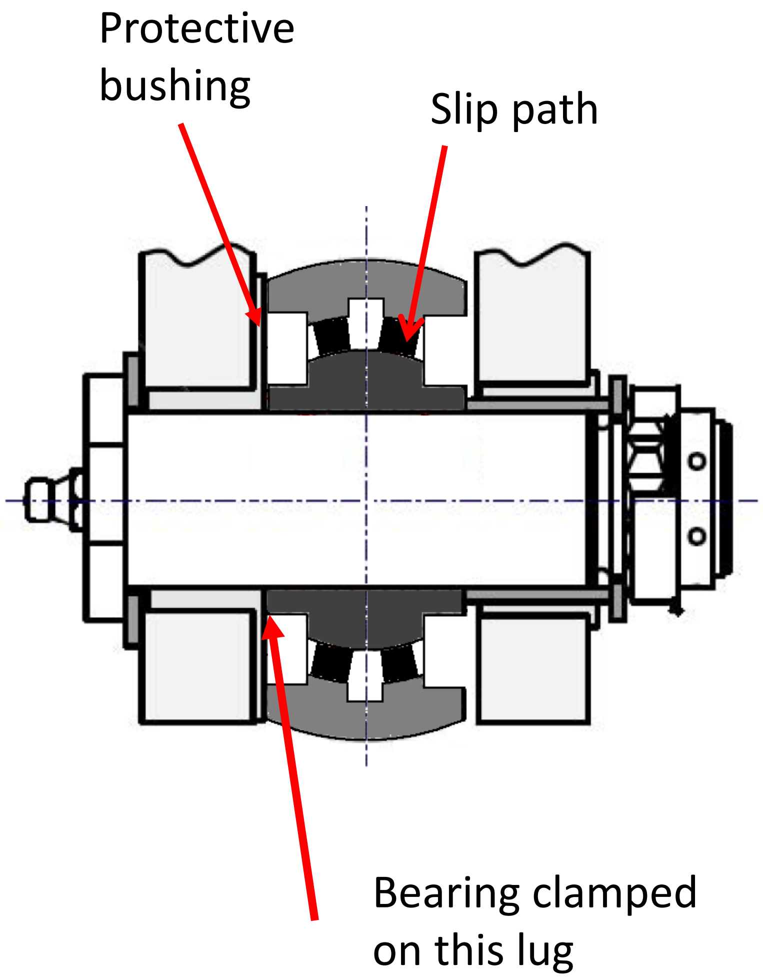


Bearings – Installation - Clamped Installation

A bearing is described as clamped when the inner race is prevented from rotating relative to the bolt by the friction torque applied to the bearing lateral faces. To ensure that the rotation occurs between the inner and outer ring of a bearing (typically the primary slip path), and not between the inner ring and the bolt, the inner ring of the bearing is locked against rotation. This can be achieved through applying a sufficient bolt pre tension load or the use of a dedicated locking device.

A joint using a clamped bearing would have one primary slip path (bearing).

In the event of bearing seizure, the performance of the secondary slip path depends upon the magnitude of the clamping torque applied.



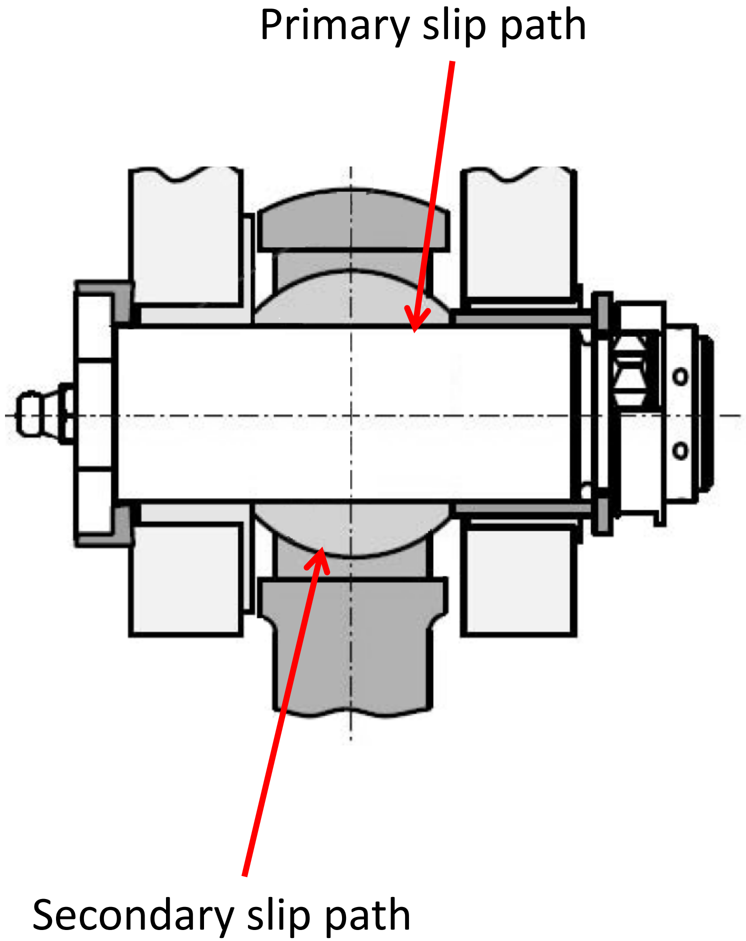
Bearings – Installation - Low Clamped Installation

A bearing is described as low-clamped when the inner race is allowed to rotate relative to the bolt by a minimal friction torque applied to the bearing lateral faces, and the function of the bearing is limited to misalignment compensation. The friction torque is only required to ensure positive lateral location of the bearing inner race.

A joint using a low-clamped bearing would have one primary slip path (bearing/bolt) and one secondary slip path (bearing).

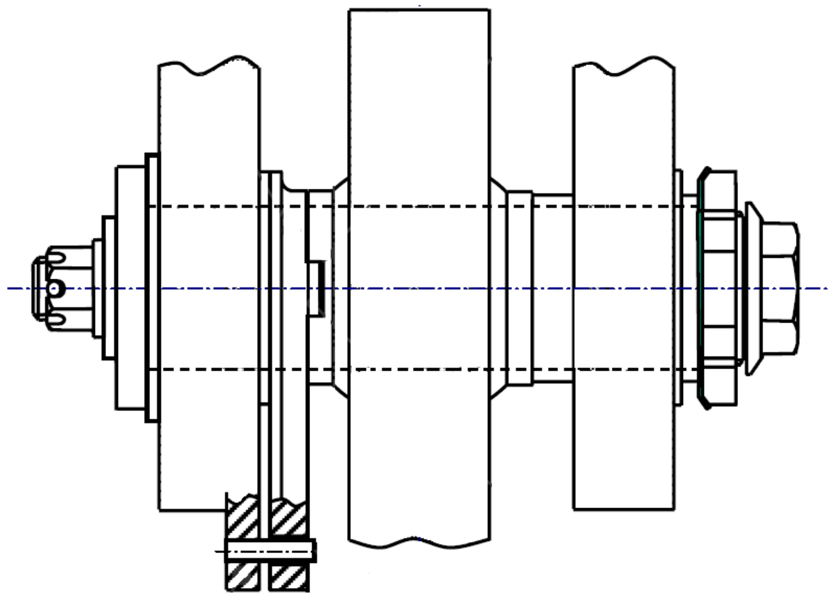
The bolt shall be installed with a retainer device to prevent its rotation

Lubrication is required



Bearings – Installation - Restrained Installation

A bearing is described as restrained when the inner race is physically prevented from rotating relative to the bolt using a form-fit locking feature between the clevis and the inner race. In the case of bearing seizure, the locking feature may be designed to fail at a pre-determined load allowing the inner race to rotate on the bolt. The clamping torque is only required to ensure positive lateral location of the bearing inner race and should therefore be as low as possible.



A joint using a restrained bearing would have one primary slip path (bearing) and one secondary slip path (bearing/bolt activated after failure of the form-fit locking feature).

Bearings - Installation

Rolling bearings

- Rolling element bearings should be installed with the inner race clamped.
- Although total seizure of rolling bearing is unlikely, if a secondary slip path between the inner race of the bearing and the bolt is required bolt should be restrained with an anti-rotation device and periodically lubricated with grease.

Spherical bearings self-lubricated

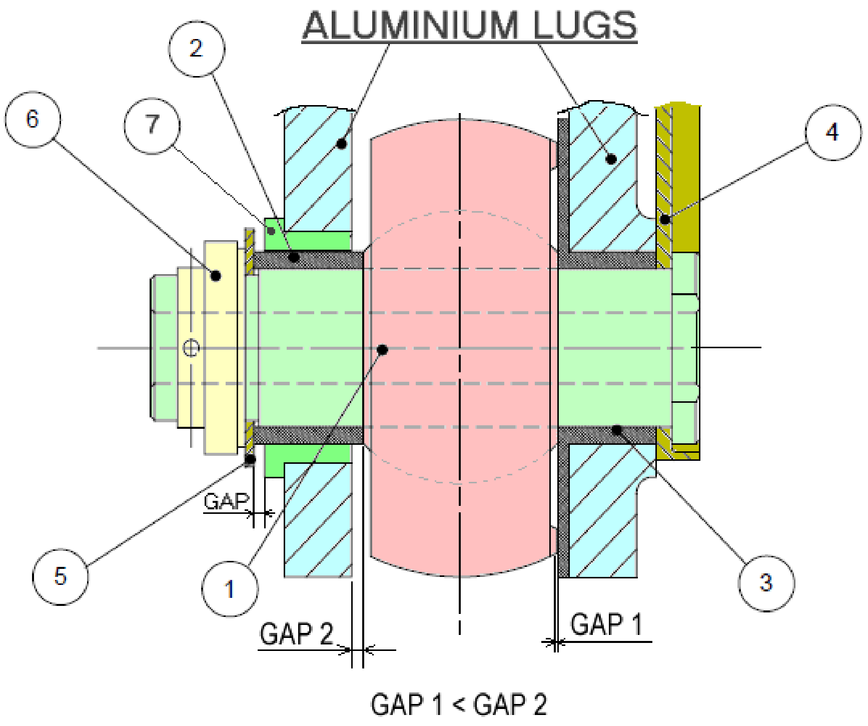
- Depending on the installation, spherical bearings should be installed clamped, low clamped or restrained
- Total seizure of spherical bearings cannot be regarded as extremely improbable and therefore a secondary slip path should be provided between the inner race and the bolt, which should be treated with low friction coating and anti-rotation device.

Bearings - Installation

Spherical metal to metal bearings

- Spherical bearings should be installed clamped or low clamped.
- Spherical metal to metal bearings lubricated with grease have a moderate coefficient of friction. For clamped installations, clamping torques tend therefore to be higher than rolling element bearings, but lower than spherical self-lubricated bearings.
- Total seizure of spherical bearings cannot be regarded as extremely improbable and therefore a secondary slip path should be provided between the inner race and the bolt, which should be restrained with an anti-rotation device and periodically lubricated with grease.

Bearings Installation examples



CLAMPED ROLLER BEARING

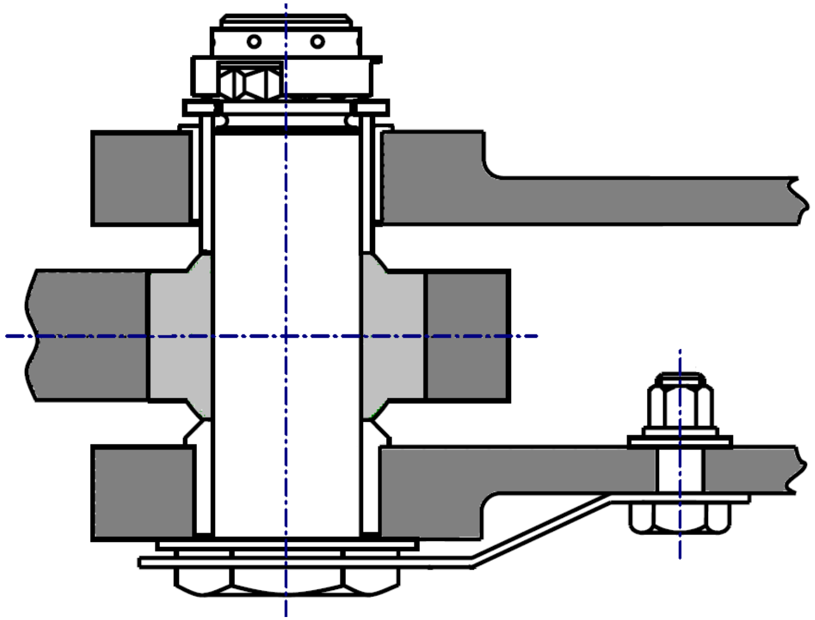
Item	Material	Description
1	CRES	Bolt; special coating on outer dia
2	CuBe	Plain bush; sliding on bolt and clevis lug
3	CuBe	Shouldered bush; interference fit on clevis lug; achieves rod end anti-rot.
4	CRES	Bolt anti-rotation device
5	CRES	Washer
6	CRES	Nut
7	CuBe	Shouldered bush; interference fit

Bearings – Anti-rotation devices

There are two basic types of device:

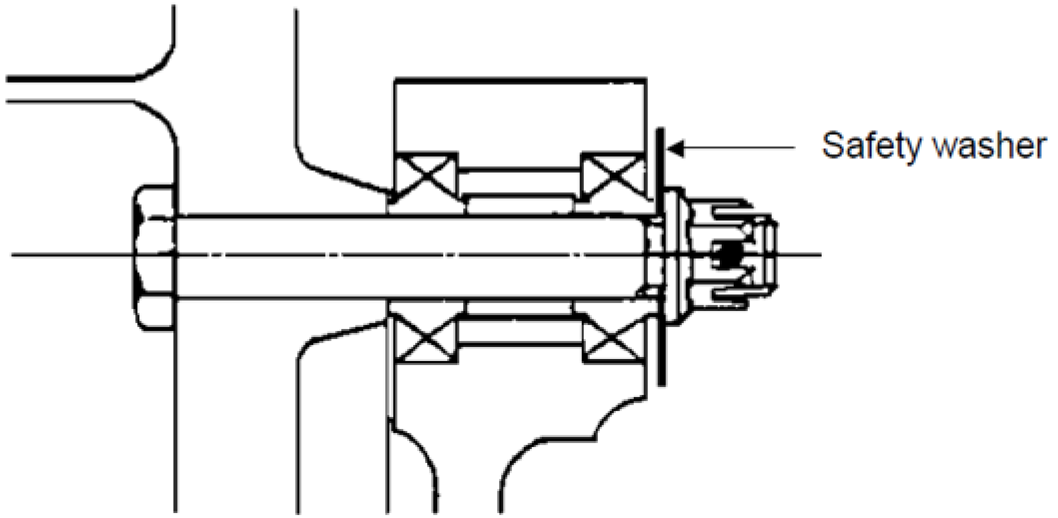
- Structural feature
- Separate Anti-rotation device (preferred solution)

The anti-rotation device is a separate item fitted under or over the head of the bolt



Bearings – Cantilever installation

A sufficiently wide safety washer shall be incorporated in the installation to retain the bellcrank, pulley or rod end to prevent the bush or, spherical bearing become loose.



Control cables

Control cables will be in accordance with the standard MIL- W-83420 and must meet the requirements stated in the regulation CS 25.689:

(a) Each cable, cable fitting, turnbuckle, splice, and pulley must be approved. In addition –

(1) No cable smaller than 3.2 mm (0·125 inch) diameter may be used in the aileron, elevator, or rudder systems; and

(2) Each cable system must be designed so that there will be no hazardous change in cable tension throughout the range of travel under operating conditions and temperature variations.

(b) Each kind and size of pulley must correspond to the cable with which it is used. Pulleys and sprockets must have closely fitted guards to prevent the cables and chains from being displaced or fouled. Each pulley must lie in the plane passing through the cable so that the cable does not rub against the pulley flange.

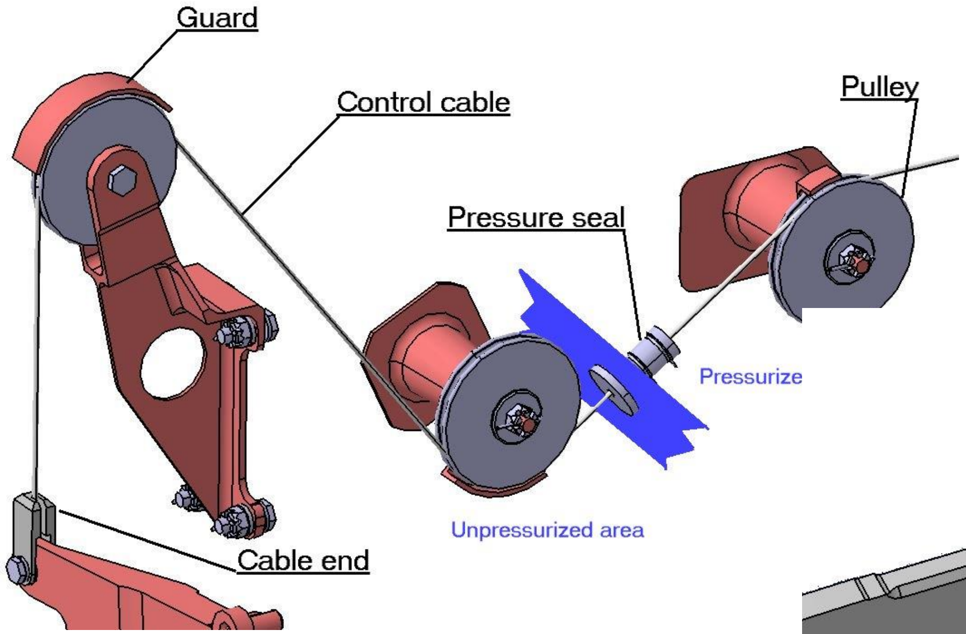
(c) Fairleads must be installed so that they do not cause a change in cable direction of more than three degrees.

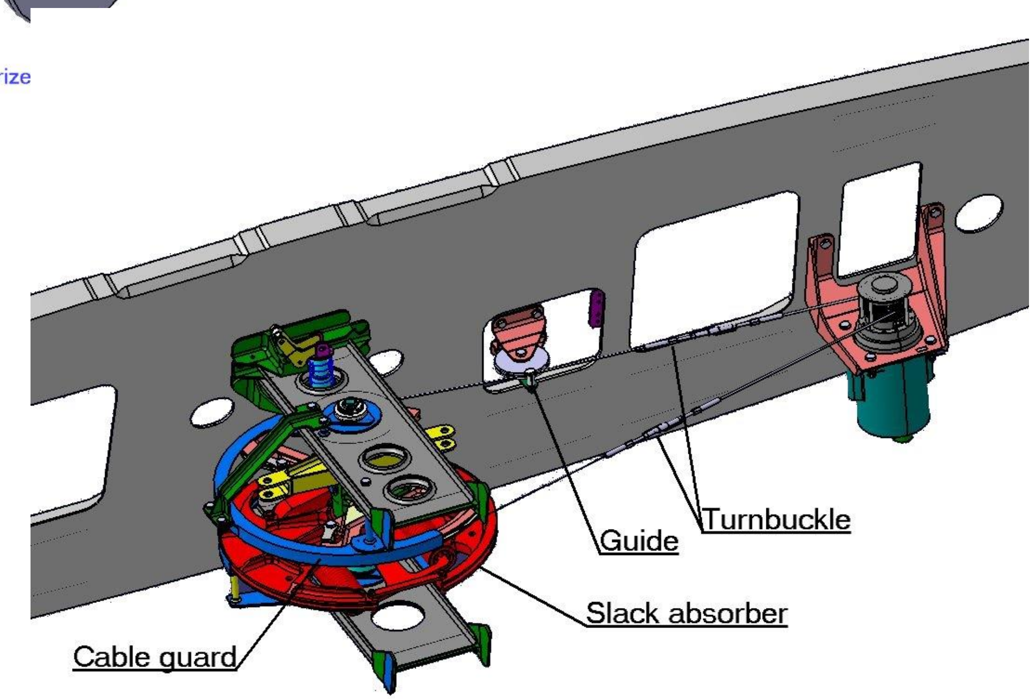
(d) Clevis pins subject to load or motion and retained only by cotter pins may not be used in the control system.

(e) Turnbuckles must be attached to parts having angular motion in a manner that will positively prevent binding throughout the range of travel.

(f) There must be provisions for visual inspection of fairleads, pulleys, terminals, and turnbuckles.

Control cables – General view

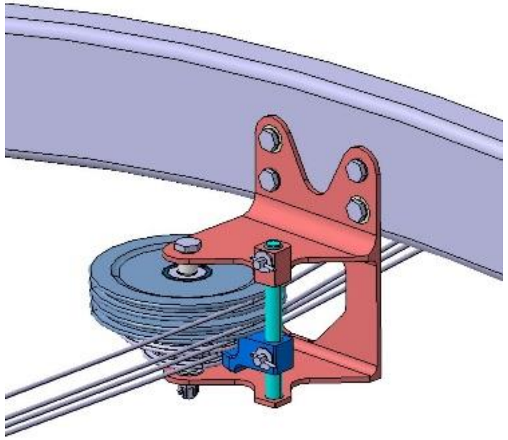
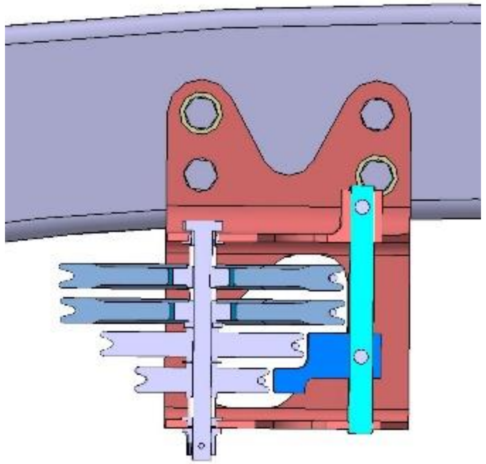




Control cables

Cables fatigue life and wear is affected directly by the number of sheaves; so, pulleys spacing is a compromise between ensuring the cable does not come out the groove and minimizing its wear

Maximum recommended distance between:	Pulleys (mm)	Guides (mm)
Fuselage	5000	1500
Wing	2000	1000

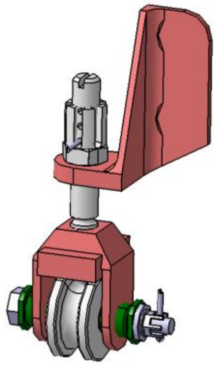



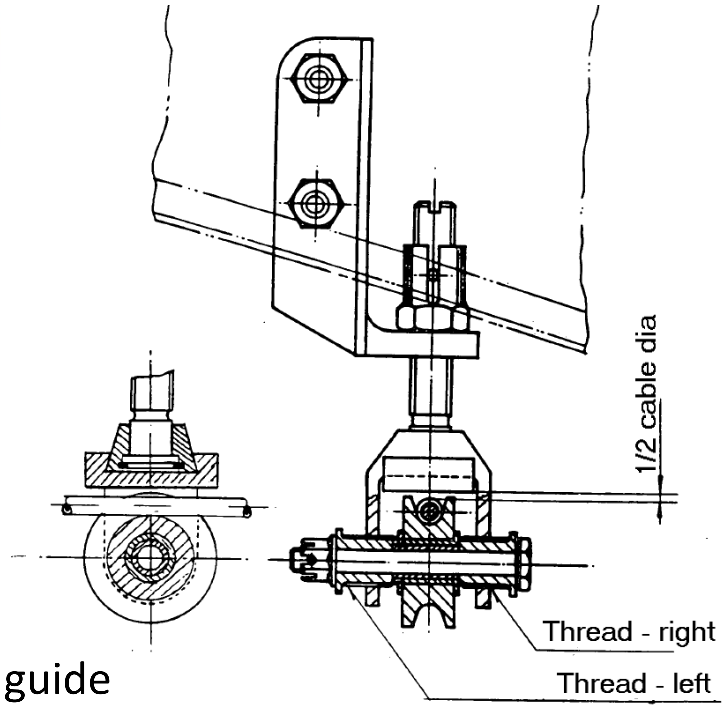
Pulleys

Component installation requirements

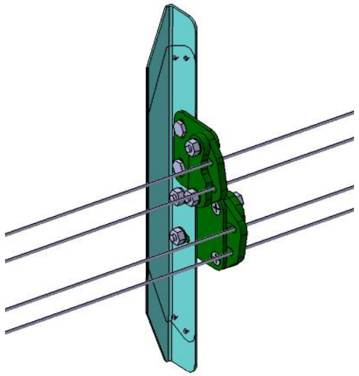
Control cables

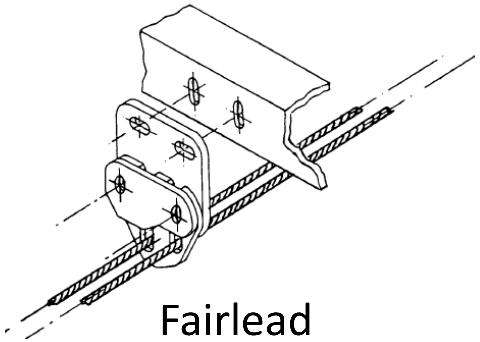
- Guides are used to avoid cable displacements, to control low clearance areas and to avoid incorrect assemblies
- The cable does not necessarily have to be in contact with the guides
- Rigging capability is needed





Cable guide





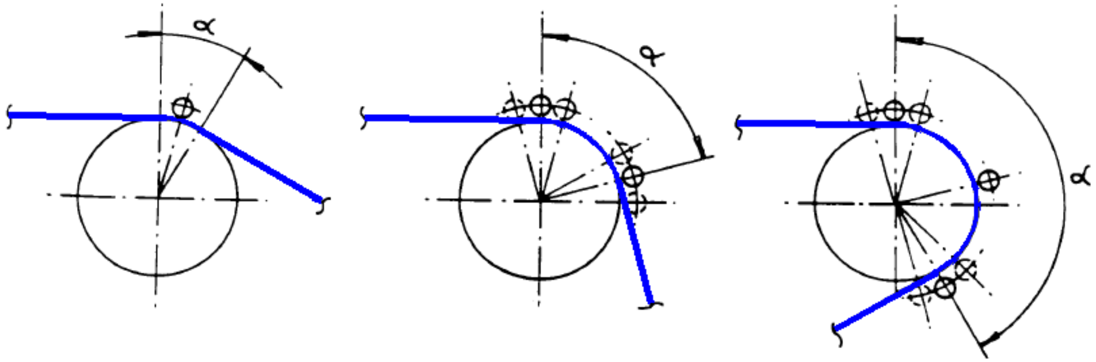
Fairlead

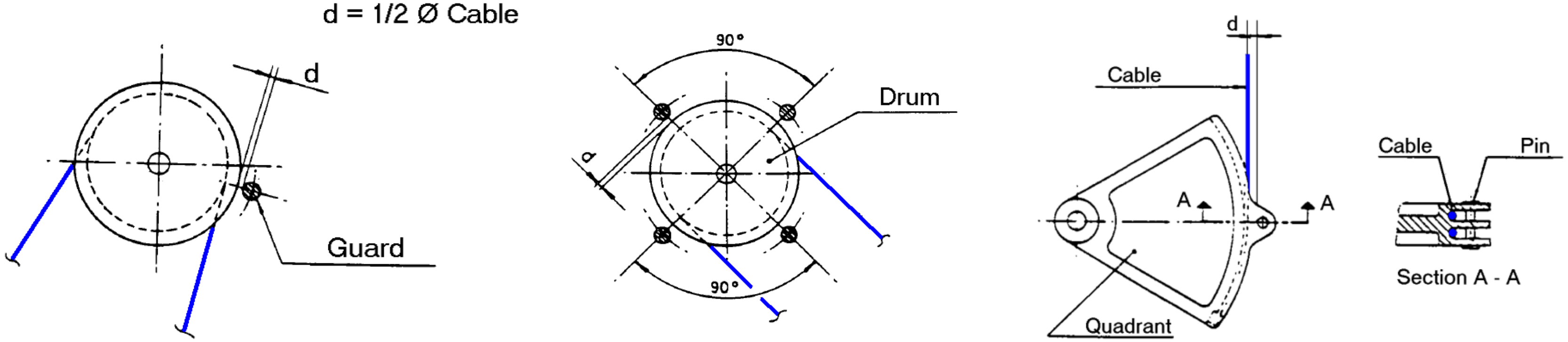
Component installation requirements

Control cables

Pulley guards: number and location

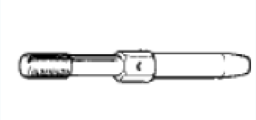
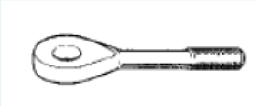
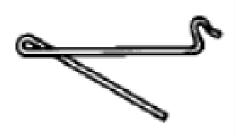
α	Guards
Up to 30°	1
30° to 90°	2
90° to 180°	3

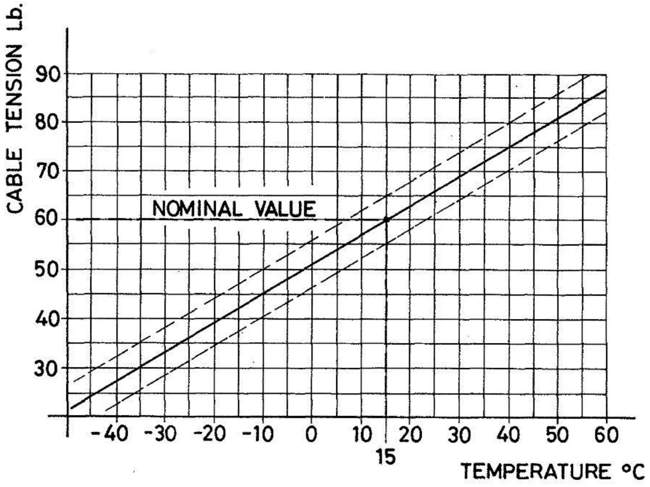


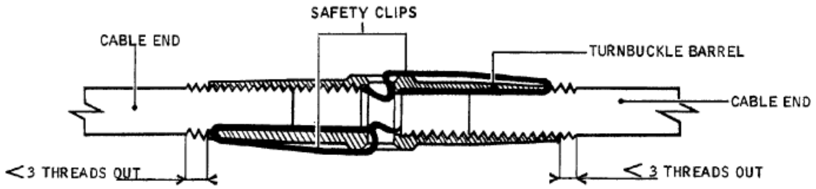


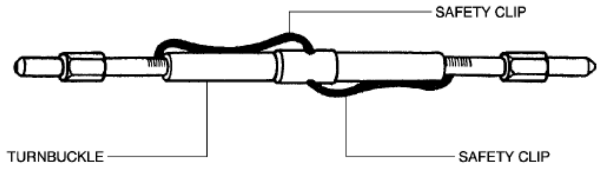
Control cables - Turnbuckles

- Turnbuckles are used to joint different cable sections and to apply rigging tension to the cable system
- Rigging load depends on temperature

Description	Standard	Representation
Turnbuckle body, clip locking	MS21251	
Terminal, wire rope, swaging stud	MS21254	
Eye end, turnbuckle, clip locking	MS20668	
Clip locking	MS21256	

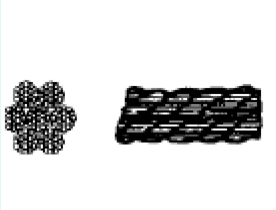
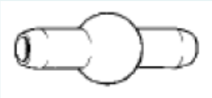
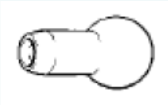
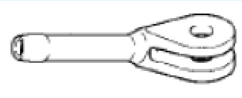
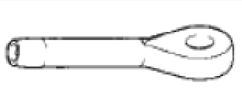






Control cables - Cables

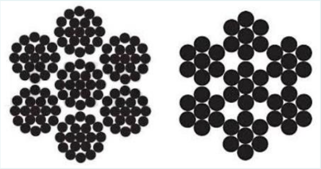
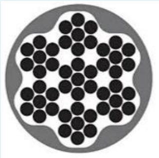
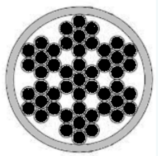
— Cable length to be obtained after proof loading (60% minimum breaking load)

Description	Standard	Representation
Wire rope, flexible, Composition A	MIL-DTL-83420/1	 (1)
Wire rope, flexible, Composition B	MIL-DTL-83420/2	
Ball end, wire rope, swaging, double shank	MS20663	
Ball end, wire rope, swaging, double shank	MS20664	
Terminal wire rope, swaging, fork end	MS20667	
Terminal wire rope, swaging, eye end	MS20668	

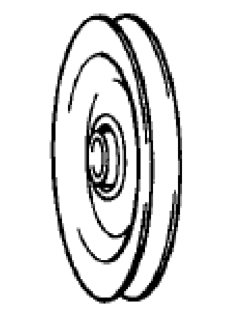
(1) Cable construction: use 7 x 19 type preferably (7 x 7 has lower fatigue properties)

Control cables per MIL-DTL-83420

- CS 25 regulations imposes no cable smaller than 3,2mm for primary flight controls
- Carbon steel cables required external lubrication to improve fatigue and corrosion protection; internal lubrication is provided by the cable supplier
- Corrosion resistant steel cables and jacketed cables do not require external lubrication

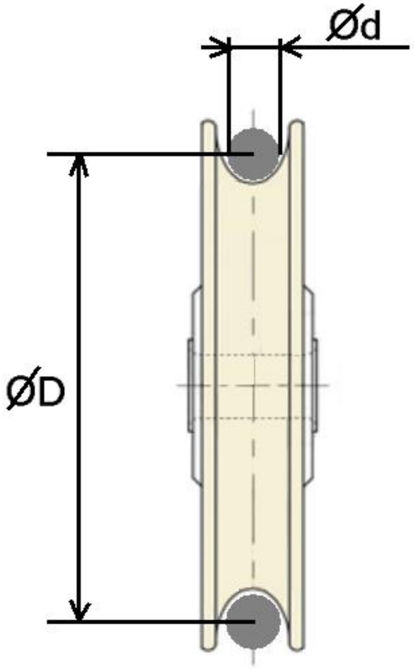
Specification	Arrange	Remarks	Representation
Type I Composition A	7 x 7 7 x 19	Carbon steel Requires external lubrication	
Type I Composition B		Corrosion resistant steel	
Type II Composition A or B		Cable Type I plus nylon jacketed Better wear and corrosion resistant No relubrication is required	
Lock-clad per MIL-C-87218		Aluminum tube swaged partially over MIL-DTL-83420 cable; clad sections can not be bended; only for straight sections	

Control cables - Pulleys

Description	Standard	Representation
Pulley groove for flight controls	MIL-DTL-7034/2 (MS20220)	
Pulley groove for secondary controls	MIL-DTL-7034/1 (MS20219)	

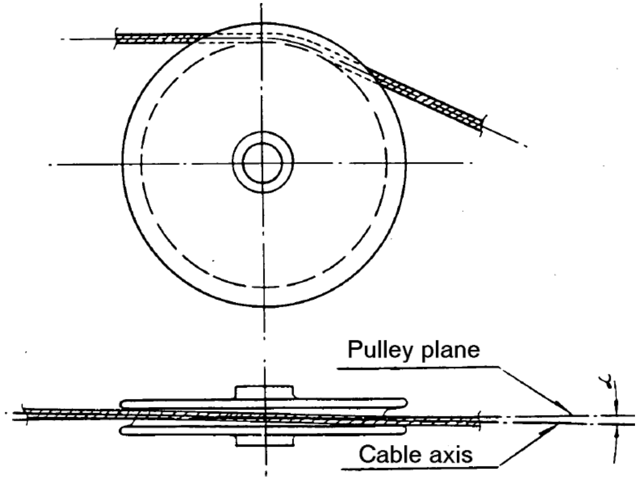
D/d	Design criteria
60	Desirable target
40	Minimum for cable life
25	Minimum (impact on friction and life)

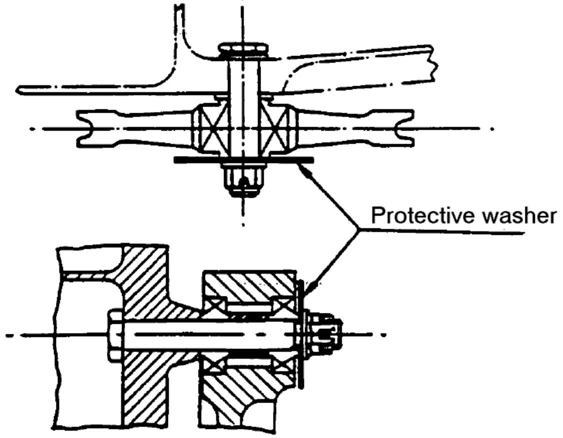
Recommended pulley groove radius = d/2 + 0,3 ± 0,1 mm



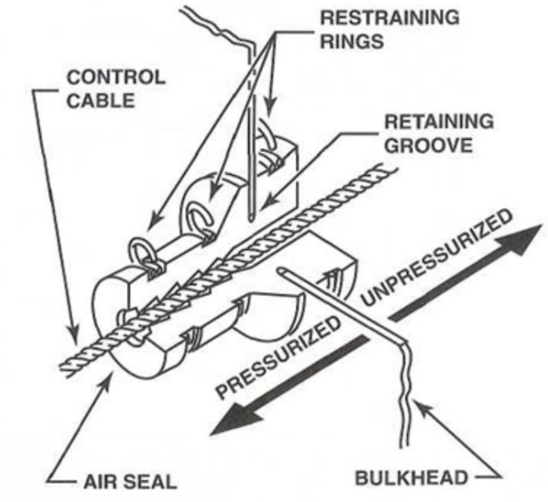
Design target is to have cero misalignment, although 0,25 degrees should be acceptable.

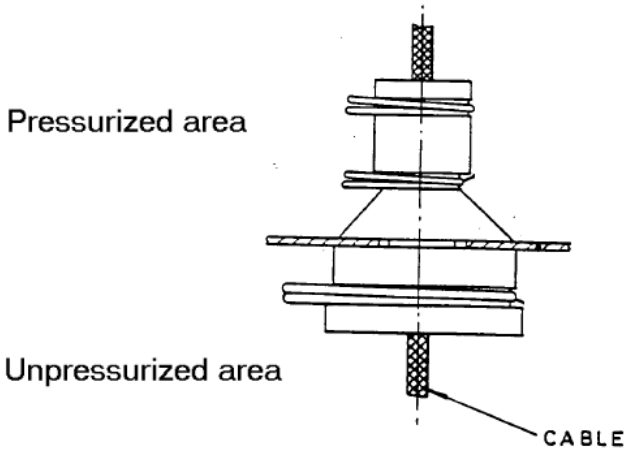
On installation, with manufacturing tolerances, 2 degrees maximum acceptable deviation

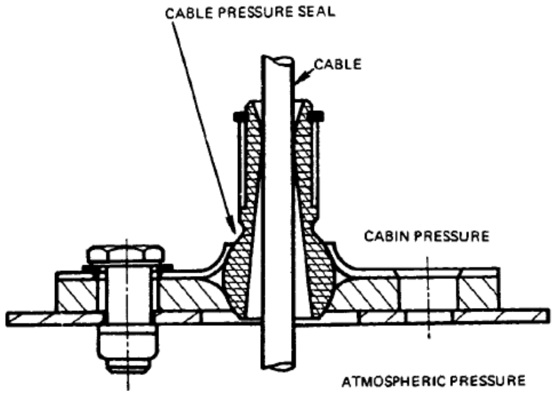


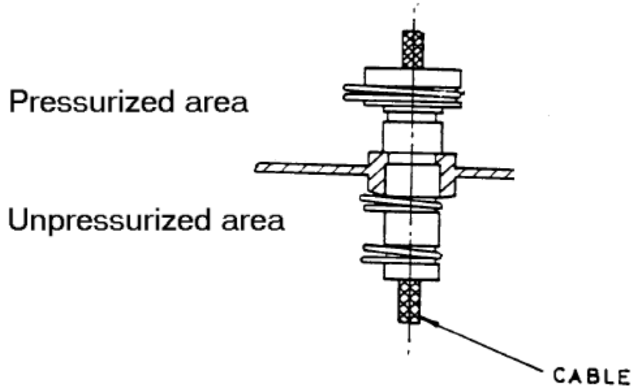


Control cables – Pressure seals



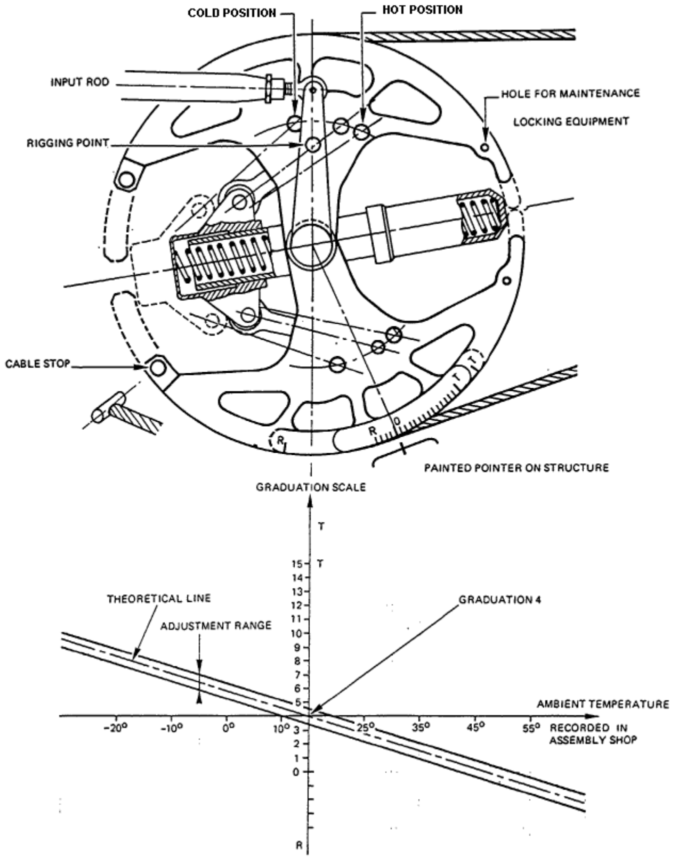






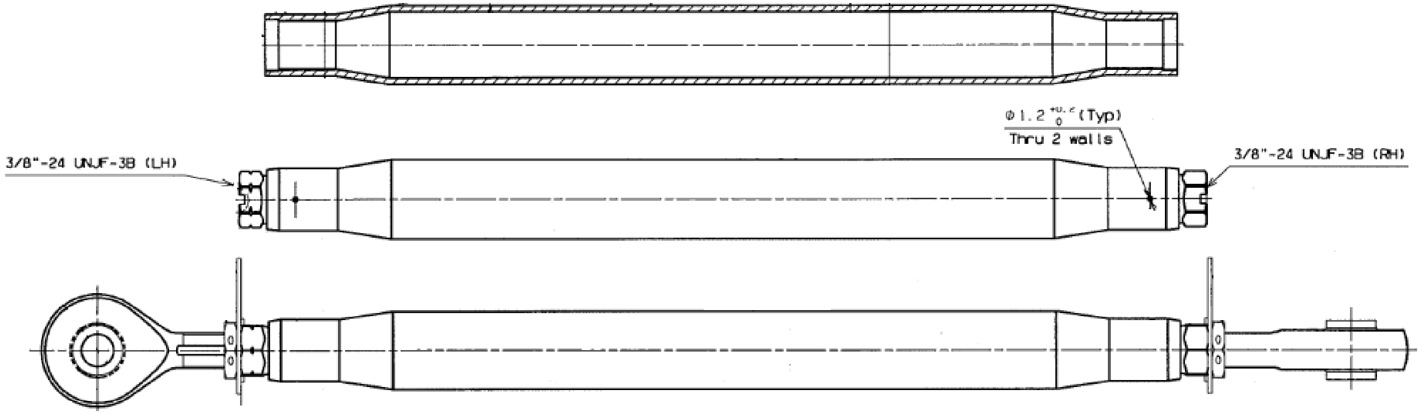
Cable tension regulator

- Cable tension regulators are used to reduced rig load variations due to temperature changes
- They also compensate tension differences between both cables



Control rods

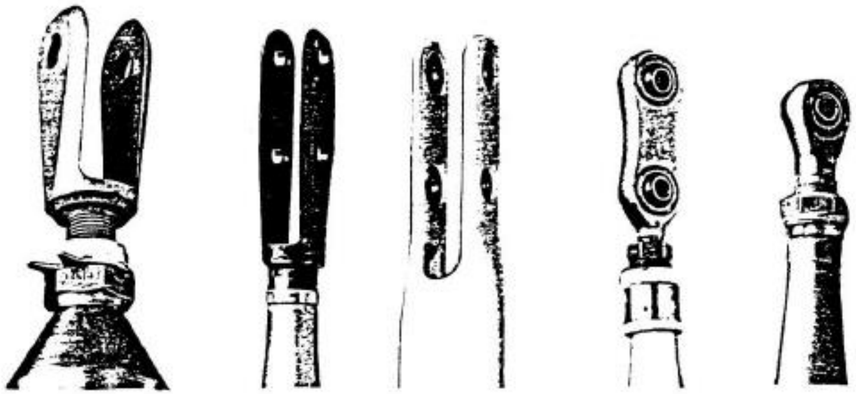
- The most efficiency solution for control rods is to have a rod body made of a thin walled tube with the ends swaged down for rod ends

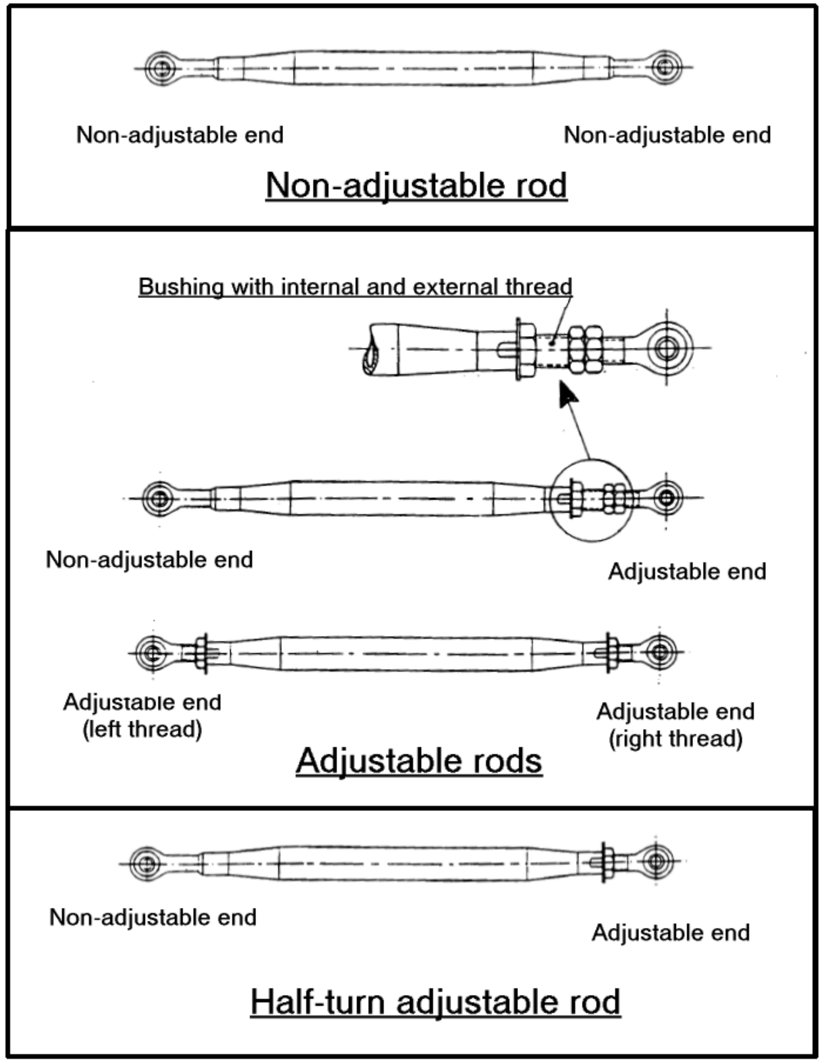


- Maximum length may be imposed by accessibility and maintainability reasons
- Drainage holes must be provided taken into account the different orientations of the rod (holes must be located at the lowest points)
- Electrical bonding should be evaluated
- Adjustable rods attached to the control surface directly, shall use steel threads

Control rods

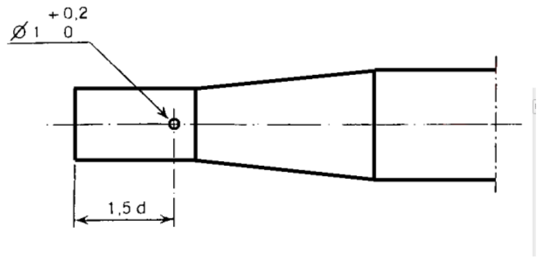
- Control rods types:
  - Fixed, non-adjustable
  - Half turn, one end adjustable
  - Continuous, both ends adjustable
  - Continuous, one end adjustable(Vernier type)
- Rod ends types:
  - Fork
  - Lug (plain or roller bearings)



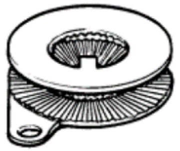


Control rods

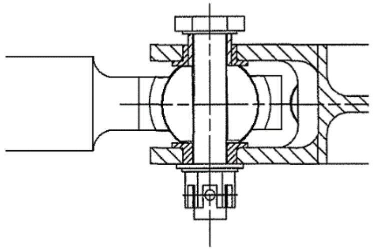
- Minimum engagement inspection hole for adjustable ends: minimum distance 1.5 x tread end diameter



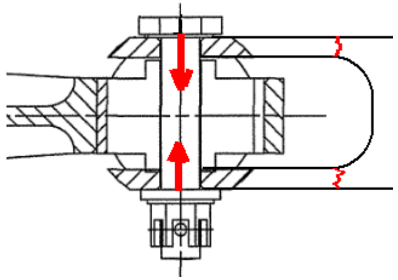
- Fixed terminals must be locking permanently



- Adjustable rod ends must be positive locking using wire lock, or special washers (e.g. NAS1193)



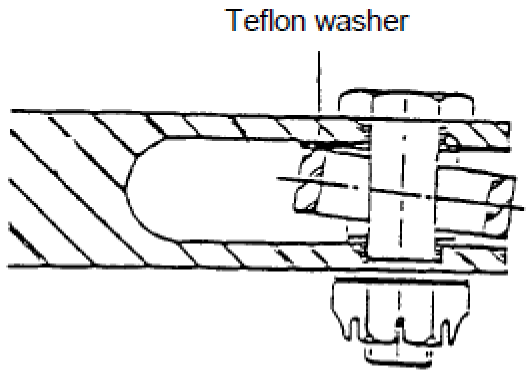
- The inner race of rod end bearing shall be clamped to avoid the rotation of the inner race around the bolt



- In case of fork ends, precautions shall be taken to avoid applying bending moment on clevis lugs

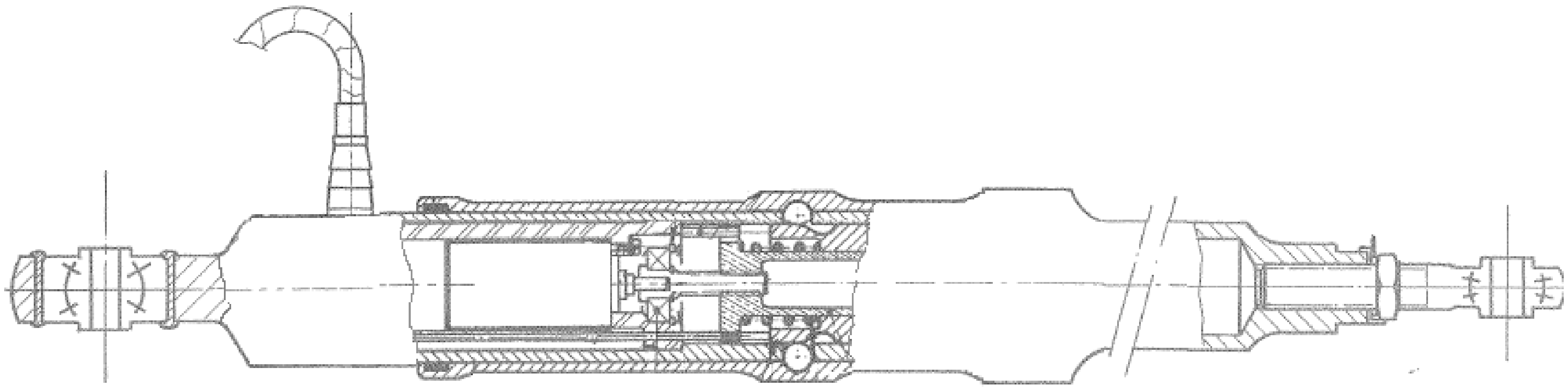
Control rods

- On rod/bellcrank attachments, precautions shall be taken to prevent sharp eye-end corners from damaging the internal protection of the fork fittings.
- To prevent such issue, anti-rotation features shall be added to the rod end and shouldered bushing shall be installed in the clevis lug; insertion of teflon washer (preferably bonded) is acceptable

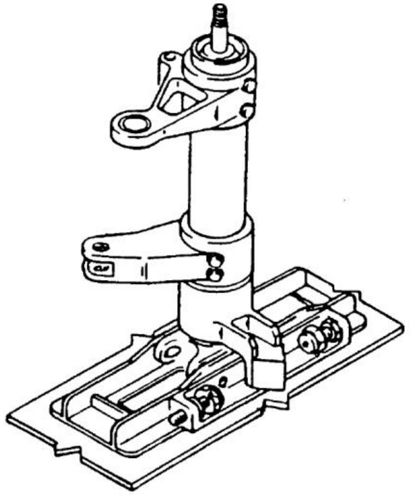


Un-coupling rod

- In the event of jamming at any point of the control system, the two control channels (pilot and copilot) can be isolated by means of un-coupling rods or equivalent devices

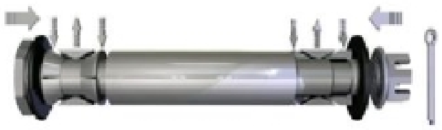


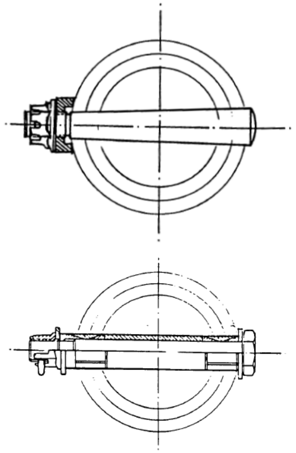
Torsion tubes



Main guidelines

- Torsional and bending stiffness are some of the main drivers
- Dimension limited by accessibility constrains
- Torsion tubes must not be common to different controls
- Tapered pins or expandable bolts shall be used to transfer loads between the different components
- Bearings installed in vertical torsion tubes must be protected against water accumulations, foreign objects or dust
- Avoid water accumulation inside the tubes (drainage shall be provided)

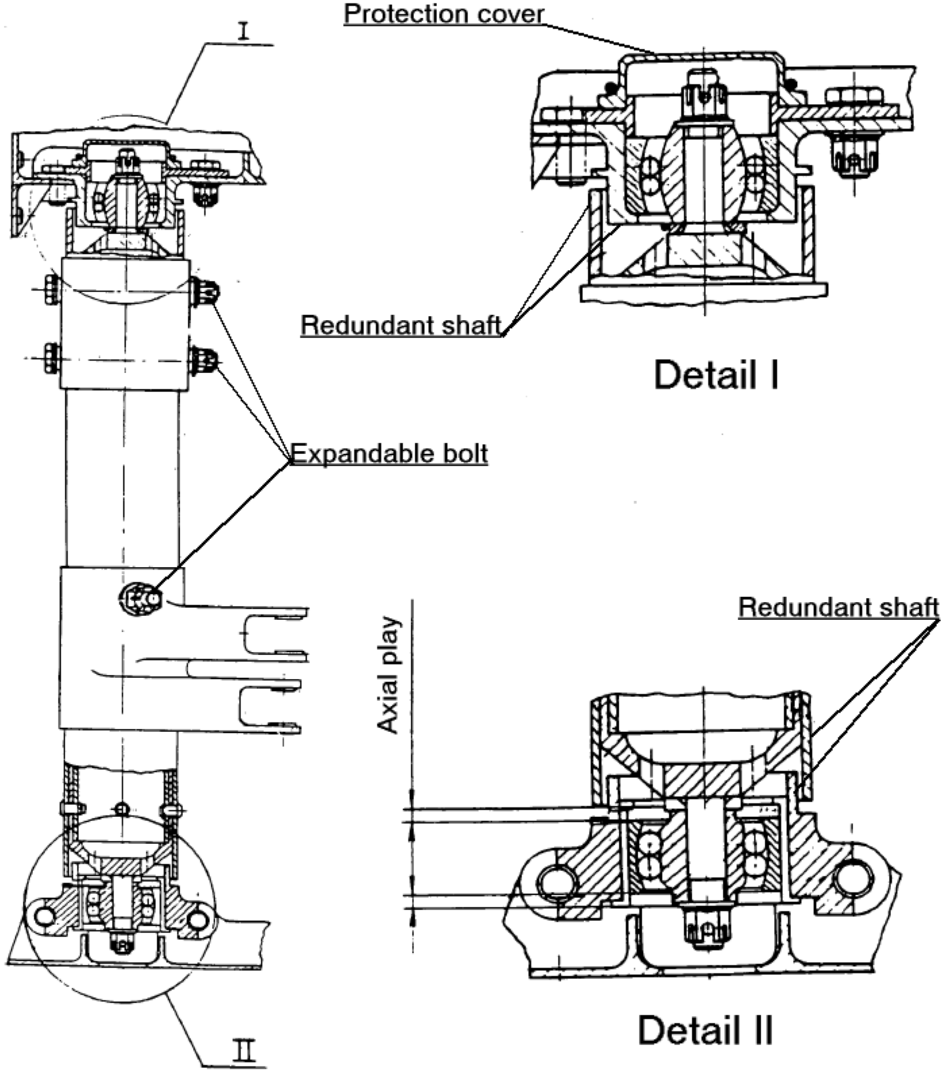




Torsion tubes

Misalignments

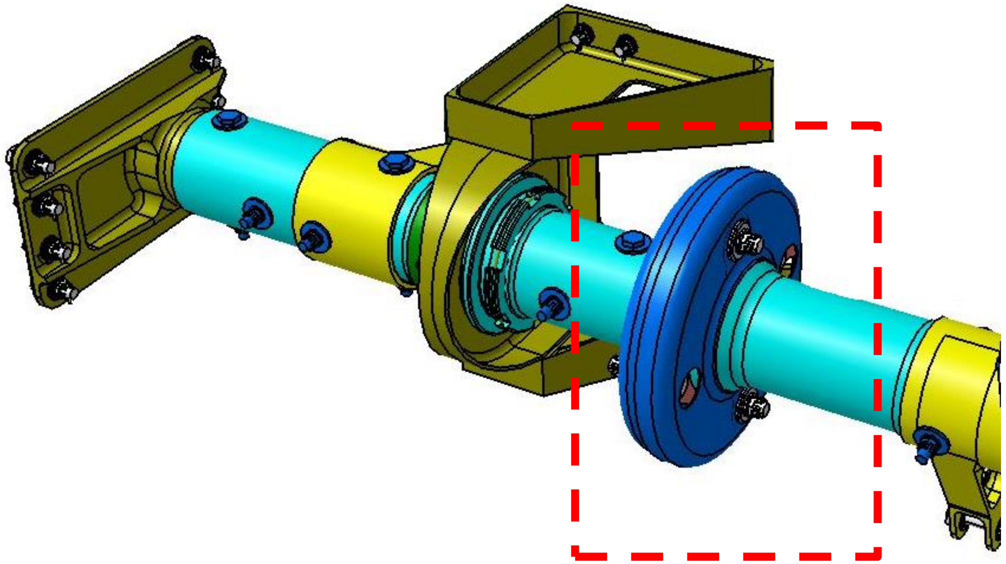
- Usually, axial loads are reacted in one end of the torsion tube while the other end has axial play
- Bearings installed on tube ends shall admit angular misalignments
- Backup devices shall be foreseen to provided redundancy in case of failure

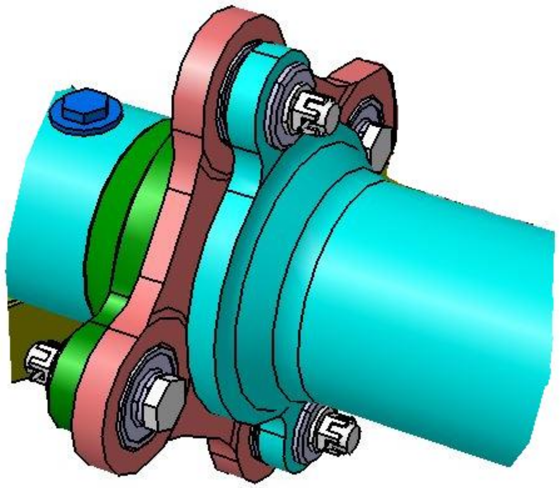


Torsion tubes

Misalignments - Angular

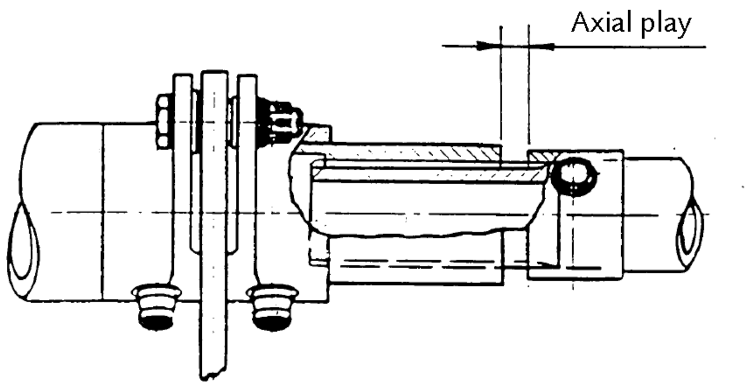
- Cardan type joints shall be used to transfer rotation between not aligned shafts or when deflections due to loads or deformations are expected



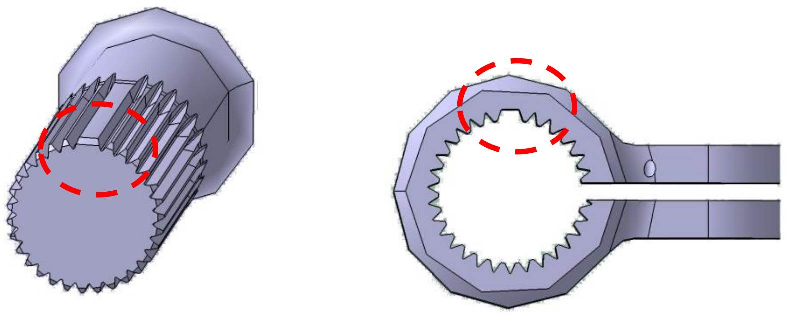


Splines

- Splines are used to transfer rotation between to shafts, when axial displacements are expected, or to joint tightly two parts (no angular misalignment is allowed)



- If precise relative orientation between both parts is required, spline shall be indexed by including a missing tooth, on one part, and a missing space on the mating part



Incorrect assembly prevention

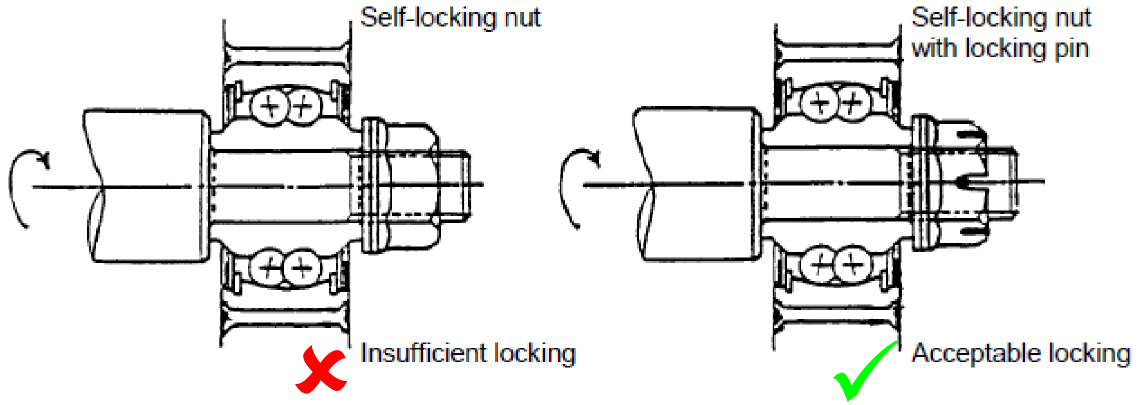
- Each element of each flight control system must be designed, or distinctively and permanently marked, to minimize the probability of incorrect assembly that could result in the malfunctioning of the system (CS 25.671 (b)).
- If several different installations of components such as rods, bellcranks, splines, torque shafts and tubes electrical connectors… are feasible, the designer shall ensure that the various installations are acceptable; installations which are not acceptable must be made physically impossible.
- Items such as control rods or shafts which, if incorrectly assembled, could lead to disconnection or incorrect kinematic function of a system should therefore be designed so that:
  1. The minimum difference in length of the items are such that it is impossible to assemble the complete system unless every item is in its correct place or,
  2. The items which could be misassembled are provided with features which render such misassembly physically impossible (e.g. projections or differently sized connection parts).

Fasteners - General

- As a general rule, connecting elements (bolts, screws, nuts, washers, etc.) must be made from corrosion resistant material.
- Tapped holes in light alloy components must be equipped with inserts.
- Screws with diameter smaller than 6,35 mm (0,25 inch) shall not be used where failure of one screw could lead to malfunctioning.
- The fasteners and their locking devices may not be adversely affected by the environmental conditions (such as temperature, vibration, shock, etc.) associated to the particular installation
- Hinge pins and attachment bolts shall be fitted head up wherever possible. If it is impossible, some suitable bolt retention method must be provided.
- If the loss of a bolt or its partial disconnection causes disconnection of a control component or a foul between the component and other linkage or the structure, the bolt shall be fitted with a retention device to cater for nut loss.
- Accessibility and maintainability requirements shall be taken into account for bolts installation and removal

Fasteners - General

- In order to limit maintenance errors, removable pin locking for similar parts in the same area shall be identical, if possible, or of the same type.
- On a bolt subjected to rotation, a self-locking nut shall be used only when a non-friction locking device is used in addition.



- Self-locking pin or bolt which, when released from its housing after becoming unlocked, would create an interference, shall have an additional non-friction locking device.
- If a bolt must be installed in a certain direction to avoid an interference with a moving part, means shall be provided ensuring that the bolt always be installed in that direction.

Fasteners - General

- Fail-safe bolts shall be used when the failure of a single bolt could lead to the control system failure.
- Permanent deformation (swaging, ball staking, etc) as locking systems shall be avoided

Fasteners – Airworthiness Regulations

- CS 25.607 Fasteners (Change 15) states:

“each removable bolt, screw, nut, pin or other removable fastener must incorporate two separate locking devices if:
• its loss could preclude continued flight and landing within the design limitations of the aeroplane using normal pilot skill and strength
• or Its loss could result in reduction in pitch, roll or yaw control capability or response below that required by CS 25 Subpart B.”

- ACJ 25.607 Fasteners (Acceptable Means of Compliance) (Amendment 16) states:

“FAA Advisory Circular AC 20-71 Dual Locking Devices on Fasteners, date 12-8-70, is accepted by the JAA as providing acceptable means of compliance with CS 25.607.”

Fasteners – Airworthiness Regulations

- AC 20-71 Dual Locking Devices on Fasteners (12/8/70) states:

“5. LOCKING DEVICE. By a locking device is meant the feature incorporated on the fastener which will prevent the loss of this fastener and retain the fastener in its proper installation. The locking devices on the fasteners may be either friction or nonfriction types. Permanent, solid, nonmetallic friction inserts are considered acceptable as friction-type locks provided they perform their intended functions,

e. Bolts which are subject to rotation must have at least one nonfriction-type lock. The other lock, if dual locks are required, may be either a friction or nonfriction-type lock.

f. Fasteners which are not subject to rotation may have the following type locks if dual locks are required:

(1) Two nonfriction-type locks.
(2) Two friction-type locks.
(3) One friction-type and one nonfriction-type lock.”

Protection

Flight controls installation must be protected against degradation due to:

- Environment:
  - Icing: The flight control components shall be designed to minimize the risks of jamming due to icing. Any cavity that collects condensation or running water shall be suitably protected or drained by means of holes; drain hole diameter shall be from 6 mm (objective) to 4 mm (minimum). The drain hole shall be drilled at the lowest point in the installed position; when a drained element may have several positions in space, as many drain holes as necessary shall be provided
  - Temperature variations
  - Pressure changes
  - Vibrations

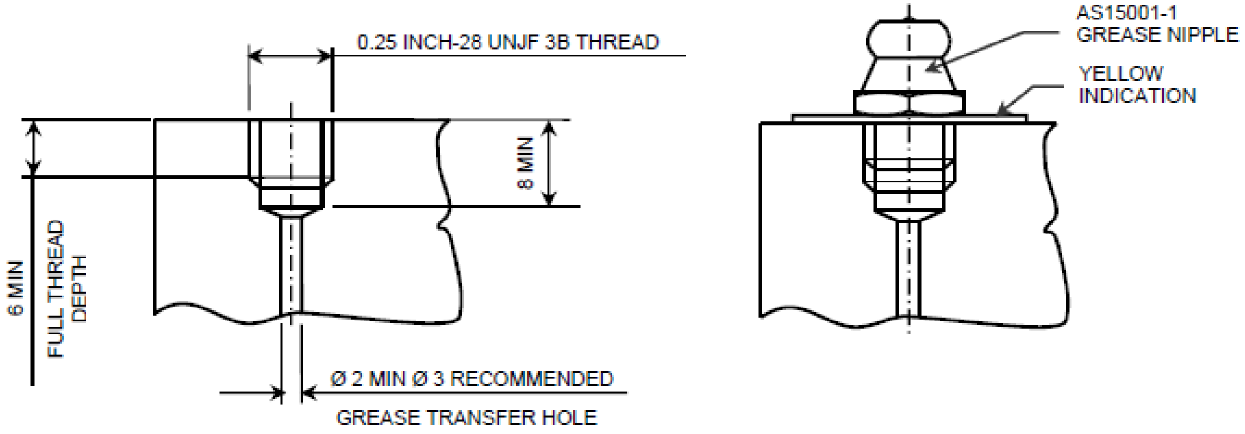
Protection

Flight controls installation must be protected against degradation due to:

- Induced factors
  - Adjacent components failures
  - FOD
  - Engine/tire burst
  - Structural failures
- Flight crew errors
  - Lock actuation of controls that must not be operated during certain flight stages
- Maintenance mistakes:
  - Adequate accessibility
  - Foolproof installations
  - Clear rigging instructions

Lubrication

- It shall be a design requirement that flight control components can be lubricated in situ except for life lubricated components
- The same lubricant should be used on all devices if possible
- Grease fittings shall be grease nipple type in accordance with AS 15001

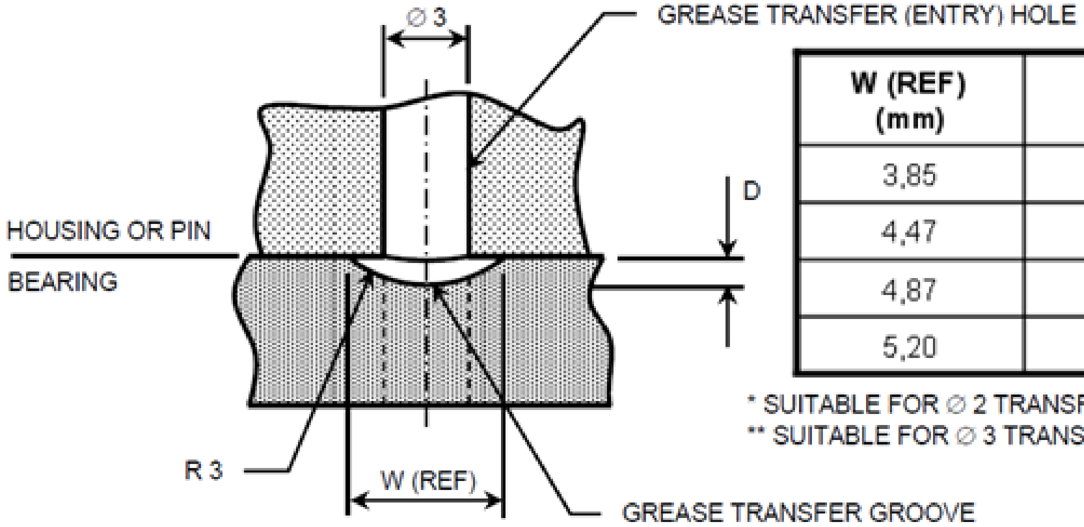


- Angled grease nipples under standard AS 15001 are available (45° and 90°), but because of it is not possible to assure their final orientation the straight ones are preferred

Assembly requirements

Lubrication

- A grease point may be situated in the bearing housing (lug) or at the end of the pin. Provision of tapped holes, grooves and drillings can all affect the stress level.
- A grease transfer groove is normally provided on the outside surface of the outer race of the bearing or the inside surface of the inner race which aligns axially with a grease transfer hole in the housing or pin.
- A grease transfer groove should have a minimum cross sectional area of 50% of the cross-sectional area of the grease transfer hole that feeds it.



W (REF) (mm)	D (mm)	Groove AREA (mm²)
3,85	0,70	1,84*
4,47	1.00	3,10
4,87	1,25	4,27**
5,20	1,50	5,53

* SUITABLE FOR ∅ 2 TRANSFER HOLE (AREA 3,14 mm²)
** SUITABLE FOR ∅ 3 TRANSFER HOLE (AREA 7mm²)

Lubrication

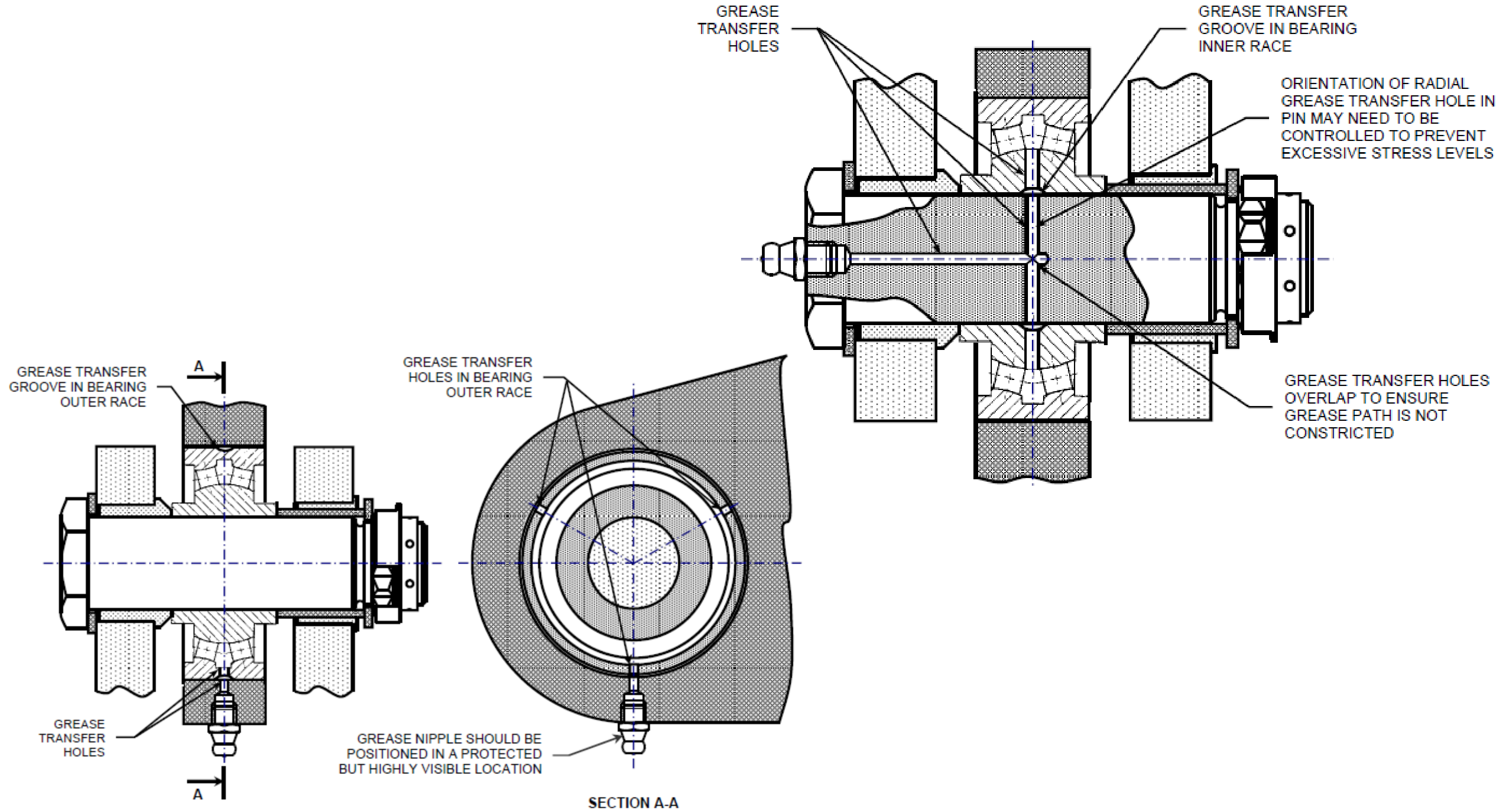
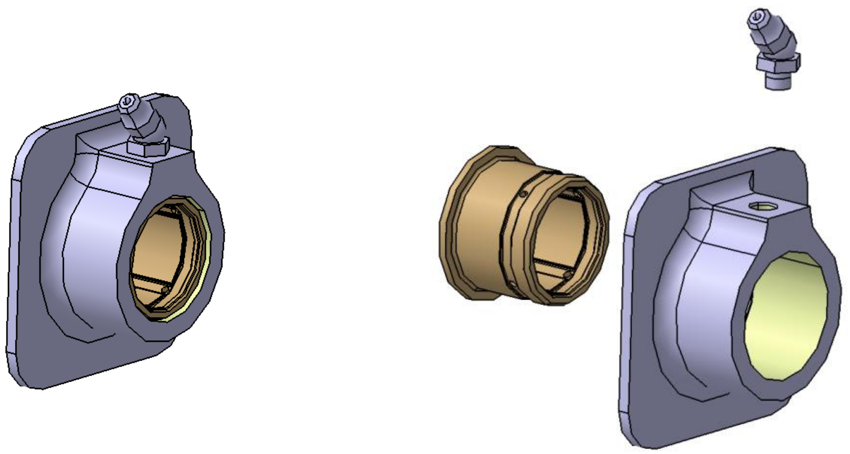


Figure 7-28: Grease Transfer Through Lug

Lubrication

- All grease passages shall be designed such that each bearing or section of a bearing is serviced by one grease nipple, greases passages shall be of minimum length possible
- The type and quantity of lubricant shall be indicated on label attached to the unit
- If a sealed ball bearing requires periodic lubrication, its installation shall be such that the maximum lubricating pressure cannot expel the bearing seals
- Commonly used grease are MIL-PRF-23827A or MIL-G-81322 but they are not compatible so they can not be mixed



Electrical bonding

- Electrical and non-electrical metallic accessories and equipment shall be individually bonded to the structure either directly or through adjoining components.
- Detachment of one end of a bonding lead shall not result in contact with any electrical equipment.
- Bonding of parts shall be achieved via dedicated bonding leads or dedicated bolts.
- Bonding leads shall be as short as possible but shall not restrict movement of moving components.
- Bonding leads shall be installed between the control surface structure and the fixed structure at all hinge positions to avoid the possibility of hinge seizure due to overheating as a result of arcing.

Assembly requirements

Maintenance

- Consideration should be given to sources of possible injury or damage to personnel or equipment, during maintenance
- Access to flight control components shall be sufficient to allow maintenance operations
- Flight controls shall be made accessible for inspection via normal access panels, over their entire length
- All the tools shall be standard tools if practicable
- It shall not be necessary to open structural parts which are normally closed (eg: tanks…) for any installation/removal operations
- Labels provided for maintenance (identification, lubrication, adjustment, …) shall be accessible for inspection by removing the inspection doors only

Maintenance

- If an item of equipment is installed in different areas, several labels shall be provided for maintenance as necessary, so that at least one of them (or one of each) is visible in each installation case
- Replacement of any position transducer must be possible without adjustment within the computer
- LRU removal and installation time: 30 minutes (design aim)
- It shall be possible to use lifting/hoisting equipment for the removal/installation of units whose weight exceeds 25

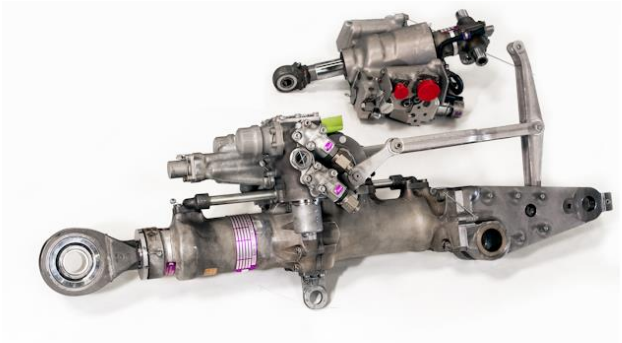
Actuators types

Control

- Mechanical
- Electrically controlled

Actuation

- Hydraulically powered
- Electrically driven
- Combined



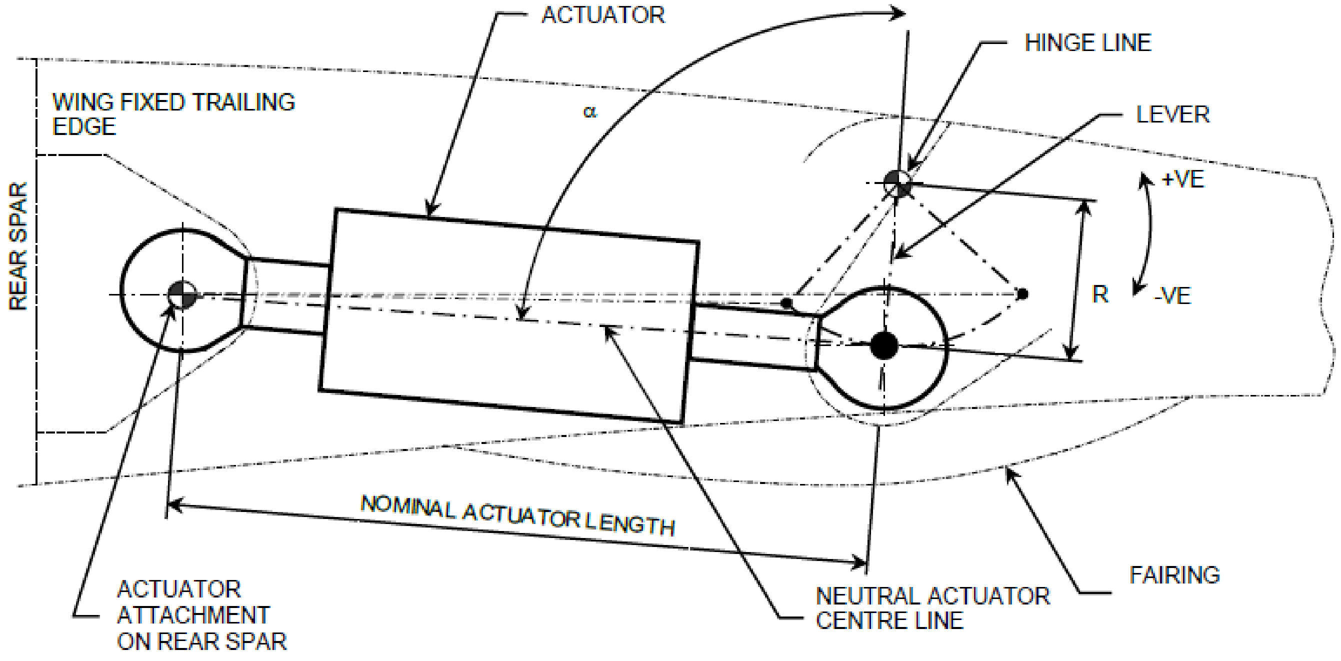
Actuators installation

Inputs

— Aerodynamic surface
  - Surface deflection
  - Hinge torque
  - Deflection speed

Output

— Actuator
  - Stroke: operational, mechanical stops
  - Functional load, Stall load
  - Actuator speed
  - Actuation radio (lever length)



Design rules and recommendations

- Actuation radio (lever length) and angle between this, at neutral position, and the actuator longitudinal axis will be defined to fulfil the control surface deflections requirements
- The design rules for the actuator attachment will be focused on
  - Increase the life of the bearing to minimize the risk of backlash source of vibrations
  - Make easy, and therefore reliable, the nut tightening torque application and nut locking
  - Minimize the risk of seizure, score, fretting and corrosion of the bolts making difficult their removability
- Based on the in service experience, it is recommended for the installation of the flight controls actuators:
  - An actuator rod end equipped with a regreasable roller bearing sealed and clamped on one clevis lug on the surface side; the rotation, primary slip path, will take place inside the bearing
  - On the rear spar side, a self lubricated or regreasable metal to metal bearing in the actuator fix part low clamped on one clevis lug; rotation will take place between the bolt an the bearing while the function of the spherical bearing is to compensate misalignments

Actuators installation

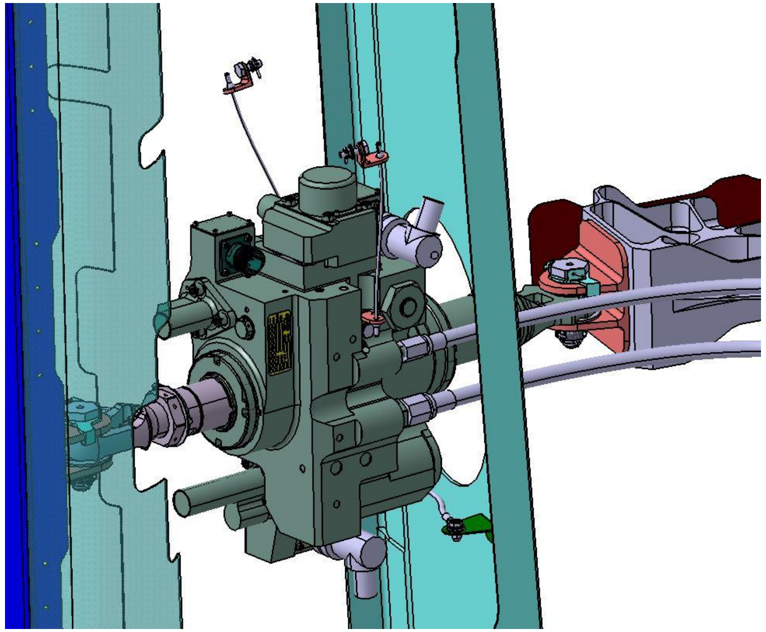
Design rules and recommendations

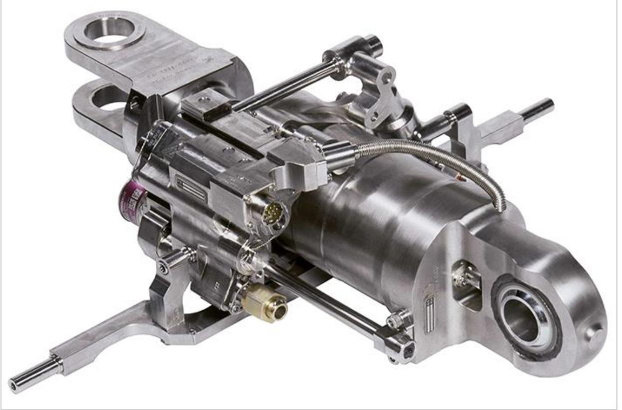
- Bolts material shall be corrosion resistant
- Clearances

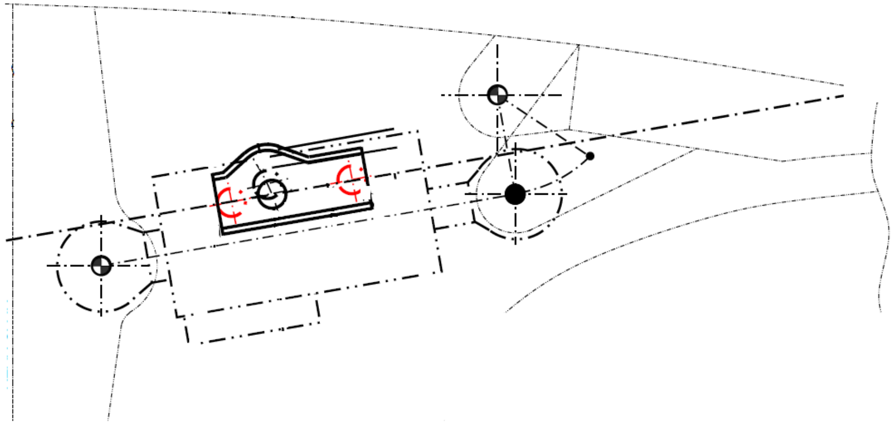
Condition		Clearance between parts 1 and 2	
Part 1	Part 2	Objective	Minimum acceptable
Actuator body	- Structure, panels or fixed equipment	25 mm	8 mm
Actuator attachment	- Fittings		4 mm locally

Jack catcher devices

- The purpose of the jack catcher is to prevent unintended contact of the actuator body with surrounding systems and structure in the event of actuator attachment failure at either end





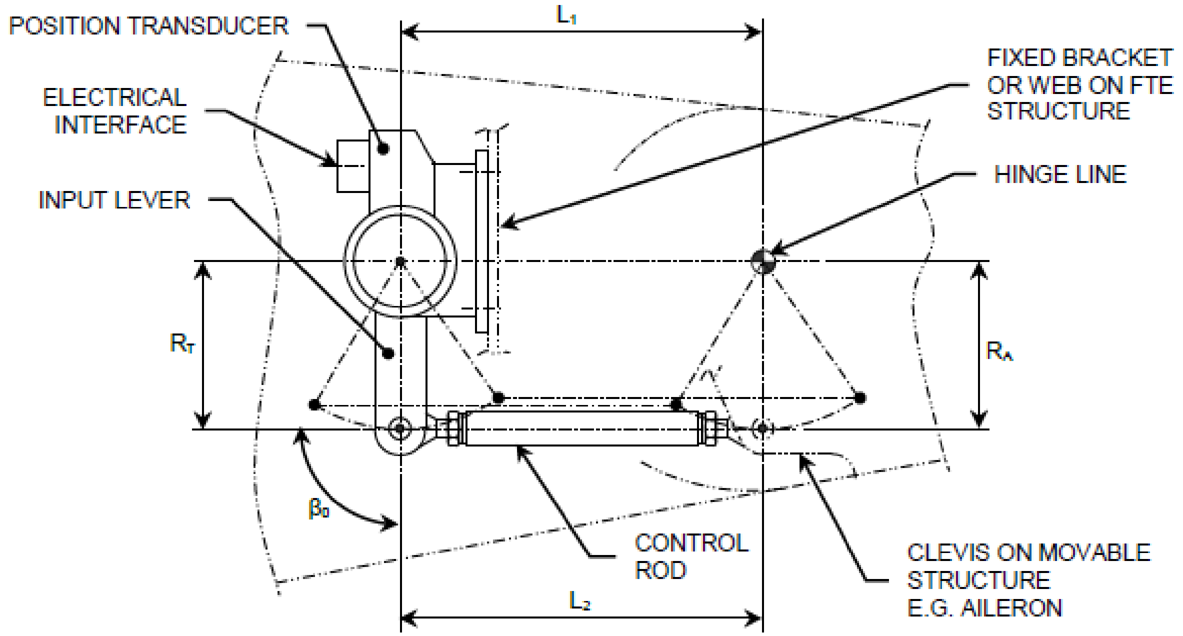


Jack catcher devices - clearances

Condition		Clearance between parts 1 and 2
Part 1	**Part 2**	**Objective**
Fixed jack catcher	Moving jack catcher	6 mm
Actuator body	Structure or panels	6 mm (in case of failure with deflections applied)
Actuator body	Adjacent systems	13 mm (in case of failure with deflections applied)

Position transducers

Position transducers are sensors measuring the mechanical position of the input controls or the movable surfaces (output sensors)



To ensure linear kinematics and a 1:1 drive ratio RT = RA and L1 = L2.

L28 References

1. Airbus System Installation Design Principles (SIDP) for Flight Controls Installations
2. Airbus Technical Design Directives for Flight Controls Installations
3. ABD 0100 (1.7) Equipment - Design - General Requirements For Suppliers
4. CS 25
5. Airbus Defence & Space Design Manual
6. SAE ARP 5770A, Design Guidelines for Aircraft Mechanical Control Systems and Components